VOLUME I: FOUNDATIONS

The Quantum Theory of Fields

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ABSTRACT: Weinberg's QFT, typeset in nicer notation. Typesetting done entirely by coding agents, mainly GPT 5.6.

Contents

Preface to Volume I

Why another book on quantum field theory? Today the student of quantum field theory can choose from among a score of excellent books, several of them quite up-to-date. Another book will be worth while only if it offers something new in content or perspective.

As to content, although this book contains a good amount of new material, I suppose the most distinctive thing about it is its generality; I have tried throughout to discuss matters in a context that is as general as possible. This is in part because quantum field theory has found applications far removed from the scene of its old successes, quantum electrodynamics, but even more because I think that this generality will help to keep the important points from being submerged in the technicalities of specific theories. Of course, specific examples are frequently used to illustrate general points, examples that are chosen from contemporary particle physics or nuclear physics as well as from quantum electrodynamics.

It is, however, the perspective of this book, rather than its content, that provided my chief motivation in writing it. I aim to present quantum field theory in a manner that will give the reader the clearest possible idea of *why* this theory takes the form it does, and why in this form it does such a good job of describing the real world.

The traditional approach, since the first papers of Heisenberg and Pauli on general quantum field theory, has been to take the existence of fields for granted, relying for justification on our experience with electromagnetism, and 'quantize' them — that is, apply to various simple field theories the rules of canonical quantization or path integration. Some of this traditional approach will be found here in the historical introduction presented in Chapter 1. This is certainly a way of getting rapidly into the subject, but it seems to me that it leaves the reflective reader with too many unanswered questions. Why should we believe in the rules of canonical quantization or path integration? Why should we adopt the simple field equations and Lagrangians that are found in the literature? For that matter, why have fields at all? It does not seem satisfactory to me to appeal to experience; after all, our purpose in theoretical physics is not just to describe the world as we find it, but to explain — in terms of a few fundamental principles — why the world is the way it is.

The point of view of this book is that quantum field theory is the way it is because (aside from theories like string theory that have an infinite number of particle types) it is the only way to reconcile the principles of quantum mechanics (including the cluster decomposition property) with those of special relativity. This is a point of view I have held for many years, but it is also one that has become newly appropriate. We have learned in recent years to think of our successful quantum field theories, including quantum electrodynamics, as 'effective field theories,' low-energy approximations to a deeper theory that may not even be a field theory, but something different like a string theory. On this basis, the reason that quantum field theories describe physics at accessible energies is that *any* relativistic quantum theory will look at sufficiently low energy like a quantum field theory. It is therefore important to understand the rationale for quantum field theory in terms of the principles of relativity and quantum mechanics. Also, we think differently now about some of the problems of quantum field theories, such as non-renormalizability and 'triviality,' that used to bother us when we thought of these theories as truly fundamental, and the discussions here will reflect these changes. This is intended to be a book on quantum field theory for the era of effective field theories.

The most immediate and certain consequences of relativity and quantum mechanics are the properties of particle states, so here particles come first — they are introduced in Chapter 2 as ingredients in the representation of the inhomogeneous Lorentz group in the Hilbert space of quantum mechanics. Chapter 3 provides a framework for addressing the fundamental dynamical question: given a state that in the distant past looks like a certain collection of free particles, what will it look like in the future? Knowing the generator of time-translations, the Hamiltonian, we can answer this question through the perturbative expansion for the array of transition amplitudes known as the S-matrix. In Chapter 4 the principle of cluster decomposition is invoked to describe how the generator of time-translations, the Hamiltonian, is to be constructed from creation and annihilation operators. Then in Chapter 5 we return to Lorentz invariance, and show that it requires these creation and annihilation operators to be grouped together in causal quantum fields. As a spin-off, we deduce the CPT theorem and the connection between spin and statistics. The formalism is used in Chapter 6 to derive the Feynman rules for calculating the S-matrix.

It is not until Chapter 7 that we come to Lagrangians and the canonical formalism. The rationale here for introducing them is not that they have proved useful elsewhere in physics (never a very satisfying explanation) but rather that this formalism makes it easy to choose interaction Hamiltonians for which the S-matrix satisfies various assumed symmetries. In particular, the Lorentz invariance of the Lagrangian density ensures the existence of a set of ten operators that satisfy the algebra of the Poincaré group and, as we show in Chapter 3, this is the key condition that we need to prove the Lorentz invariance of the S-matrix. Quantum electrodynamics finally appears in Chapter 8. Path integration is introduced in Chapter 9, and used to justify some of the hand-waving in Chapter 8 regarding the Feynman rules for quantum electrodynamics. This is a somewhat later introduction of path integrals than is fashionable these days, but it seems to me that although path integration is by far the best way of rapidly deriving Feynman rules from a given Lagrangian, it rather obscures the quantum mechanical reasons underlying these calculations.

Volume I concludes with a series of chapters, 10–14, that provide an introduction to the calculation of radiative corrections, involving loop graphs, in general field theories. Here too the arrangement is a bit unusual; we start with a chapter on non-perturbative methods, in part because the results we obtain help us to understand the necessity for field and mass renormalization, without regard to whether the theory contains infinities or not. Chapter 11 presents the classic one-loop calculations of quantum chromodynamics, both as an opportunity to explain useful calculational techniques (Feynman parameters, Wick rotation, dimensional and Pauli–Villars regularization), and also as a concrete example of renormalization in action. The experience gained in Chapter 11 is extended to all orders and general theories in Chapter 12, which also describes the modern view of non-renormalizability that is appropriate to effective field theories. Chapter 13 is a digression on the special problems raised by massless particles of low energy or parallel momenta. The Dirac equation for an electron in an external electromagnetic field, which historically appeared almost at the very start of relativistic quantum mechanics, is not seen here until Chapter 14, on bound state problems, because this equation should not be viewed (as Dirac did) as a relativistic version of the Schrödinger equation, but rather as an approximation to a true relativistic quantum theory, the quantum field theory of photons and electrons. This chapter ends with a treatment of the Lamb shift, bringing the confrontation of theory and experiment up to date.

The reader may feel that some of the topics treated here, especially in Chapter 3, could more properly have been left to textbooks on nuclear or elementary particle physics. So they might, but in my experience these topics are usually either not covered or covered poorly, using specific dynamical models rather than the general principles of symmetry and quantum mechanics. I have met string theorists who have never heard of the relation between time-reversal invariance and final-state phase shifts, and nuclear theorists who do not understand why resonances are governed by the Breit–Wigner formula. So in the early chapters I have tried to err on the side of inclusion rather than exclusion.

Volume II will deal with the advances that have revived quantum field theory in recent years: non-Abelian gauge theories, the renormalization group, broken symmetries, anomalies, instantons, and so on.

I have tried to give citations both to the classic papers in the quantum theory of fields and to useful references on topics that are mentioned but not presented in detail in this book. I did not always know who was responsible for material presented here, and the mere absence of a citation should not be taken as a claim that the material presented here is original. But some of it is. I hope that I have improved on the original literature or standard textbook treatments in several places, as for instance in the proof that symmetry operators are either unitary or antiunitary; the discussion of superselection rules; the analysis of particle degeneracy associated with unconventional representations of inversions; the use of the cluster decomposition principle; the derivation of the reduction formula; the derivation of the external field approximation; and even the calculation of the Lamb shift.

I have also supplied problems for each chapter except the first. Some of these problems aim simply at providing exercise in the use of techniques described in the chapter; others are intended to suggest extensions of the results of the chapter to a wider class of theories.

In teaching quantum field theory, I have found that each of the two volumes of this book provides enough material for a one-year course for graduate students. I intended that this book should be accessible to students who are familiar with non-relativistic quantum mechanics and classical electrodynamics. I assume a basic knowledge of complex analysis and matrix algebra, but topics in group theory and topology are explained where they are introduced.

This is not a book for the student who wants immediately to begin calculating Feynman graphs in the standard model of weak, electromagnetic, and strong interactions. Nor is this a book for those who seek a higher level of mathematical rigor. Indeed, there are parts of this book whose lack of rigor will bring tears to the eyes of the mathematically inclined reader. Rather, I hope it will suit the physicists and physics students who want to understand *why* quantum field theory is the way it is, so that they will be ready for whatever new developments in physics may take us beyond our present understandings.

* * *

Much of the material in this book I learned from my interactions over the years with numerous other physicists, far too many to name here. But I must acknowledge my special intellectual debt to Sidney Coleman, and to my colleagues at the University of Texas: Arno Bohm, Luis Boya, Phil Candelas, Bryce DeWitt, Cecile DeWitt-Morette, Jacques Distler, Willy Fischler, Josh Feinberg, Joaquim Gomis, Vadim Kaplunovsky, Joe Polchinski, and Paul Shapiro. I owe thanks for help in the preparation of the historical introduction to Gerry Holton, Arthur Miller, and Sam Schweber. Thanks are also due to Alyce Wilson, who prepared the illustrations and typed the LaTeX input files until I learned how to do it, and to Terry Riley for finding countless books and articles. I am grateful to Maureen Storey and Alison Woollatt of Cambridge University Press for helping to ready this book for publication, and especially to my editor, Rufus Neal, for his friendly good advice.

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Notation

Typesetter's note: see `NOTATION.md` for the specific notation upgrades I implemented.

Latin indices $i, j, k, \ldots$ generally run over the three spatial coordinates $1, 2, 3$. Greek indices $\mu, \nu, \ldots$ run over spacetime coordinates in the order $0, 1, 2, 3$, with x^0 the time coordinate. Repeated indices are summed unless otherwise indicated.

The spacetime metric is mostly plus,

$$\eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1),$$

so an on-shell four-momentum satisfies $p^2 = -m^2$, and spacelike separations satisfy $(x - x')^2 \geq 0$. The d'Alembertian is

$$\Box = \partial_\mu \partial^\mu = \nabla^2 - \partial_t^2.$$

The Levi–Civita tensor is totally antisymmetric, with $\epsilon^{0123} = +1$.

Spatial three-vectors are bold, and a hat denotes a unit vector: $\widehat{\mathbf{v}} = \mathbf{v}/|\mathbf{v}|$. A dot denotes a time derivative.

Dirac matrices obey

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}, \qquad \gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3.$$

With this metric convention, $\beta = i\gamma^0$ and the Dirac adjoint is $\bar{u} = u^\dagger \beta$. The step function $\theta(s)$ equals 1 for $s > 0$ and 0 for $s < 0$.

Complex conjugation, transpose, and Hermitian conjugation are written A^*, A^T, and $A^\dagger$, respectively. The identity is $\mathbf{1}$. The abbreviations +H.c. and +c.c. indicate addition of the Hermitian conjugate and complex conjugate of the preceding terms.

States use Dirac notation: $|p, \sigma\rangle$ for one-particle states, $|\Omega\rangle$ for the vacuum, and $\langle \phi | \psi \rangle$ for inner products. Asymptotic states are labeled $|\alpha\rangle_{\text{in}}$ and $|\beta\rangle_{\text{out}}$. Positive- and negative-frequency field pieces are written $\psi^{(+)}$ and $\psi^{(-)}$, avoiding the original overloading of $+$ and $-$ for both field frequencies and asymptotic states.

The interaction Hamiltonian is denoted

$$H_I(t) = \int d^3x \, \mathcal{H}_I(x),$$

in place of the original bare V notation. Other operators are generally left unhatted when the context is unambiguous.

Except in Chapter 1, $\hbar = c = 1$. Throughout, $-e$ is the rationalized charge of the electron, so $\alpha = e^2/(4\pi) \simeq 1/137$. Parentheses at the end of quoted numerical data give the uncertainty in the final digits.

1 Historical Introduction

2 Relativistic Quantum Mechanics

The point of view of this book is that quantum field theory is the way it is because (with certain qualifications) this is the only way to reconcile quantum mechanics with special relativity. Therefore our first task is to study how symmetries like Lorentz invariance appear in a quantum setting.

2.1 Quantum Mechanics

First, some good news: quantum field theory is based on the same quantum mechanics that was invented by Schrödinger, Heisenberg, Pauli, Born, and others in 1925–26, and has been used ever since in atomic, molecular, nuclear, and condensed matter physics. The reader is assumed to be already familiar with quantum mechanics; this section provides only the briefest of summaries of quantum mechanics, in the generalized version of Dirac [1].
(i) Physical states are represented by rays in Hilbert space. A Hilbert space is a kind of complex vector space; that is, if $|\Phi\rangle$ and $|\Psi\rangle$ are vectors in the space (often called 'state-vectors') then so is $\xi|\Phi\rangle + \eta|\Psi\rangle$, for arbitrary complex numbers ξ, η. It has a norm[1]: for any pair of vectors there is a complex number $\langle\Phi|\Psi\rangle$, such that

$$\langle\Phi|\Psi\rangle = \langle\Psi|\Phi\rangle^*, \tag{2.1.1}$$

$$\big\langle\Phi \;\big|\; \xi_1\Psi_1 + \xi_2\Psi_2\big\rangle = \xi_1\langle\Phi|\Psi_1\rangle + \xi_2\langle\Phi|\Psi_2\rangle, \tag{2.1.2}$$

$$\big\langle\eta_1\Phi_1 + \eta_2\Phi_2 \;\big|\; \Psi\big\rangle = \eta_1^*\langle\Phi_1|\Psi\rangle + \eta_2^*\langle\Phi_2|\Psi\rangle. \tag{2.1.3}$$

The norm $\langle\Psi|\Psi\rangle$ also satisfies a positivity condition: $\langle\Psi|\Psi\rangle \geq 0$, and vanishes if and only if $|\Psi\rangle = 0$. (There are also certain technical assumptions that allow us to take limits of vectors within Hilbert space.) A ray is a set of normalized vectors (i.e., $\langle\Psi|\Psi\rangle = 1$) with $|\Psi\rangle$ and $|\Psi'\rangle$ belonging to the same ray if $|\Psi'\rangle = \xi|\Psi\rangle$, where ξ is an arbitrary complex number with $|\xi| = 1$.

(ii) Observables are represented by Hermitian operators. These are mappings $|\Psi\rangle \to A|\Psi\rangle$ of Hilbert space into itself, linear in the sense that

$$A\big(\xi|\Psi\rangle + \eta|\Phi\rangle\big) = \xi A|\Psi\rangle + \eta A|\Phi\rangle \tag{2.1.4}$$

and satisfying the reality condition $A^\dagger = A$, where for any linear operator A the adjoint $A^\dagger$ is defined by

$$\langle\Phi|A^\dagger|\Psi\rangle \equiv \langle A\Phi|\Psi\rangle = \big(\langle\Psi|A|\Phi\rangle\big)^*. \tag{2.1.5}$$

(There are also technical assumptions about the continuity of $A|\Psi\rangle$ as a function of $|\Psi\rangle$.) A state represented by a ray $\mathcal{R}$ has a definite value α for the observable represented by an operator A if vectors $|\Psi\rangle$ belonging to this ray are eigenvectors of A with eigenvalue α:

$$A|\Psi\rangle = \alpha|\Psi\rangle \qquad \text{for } |\Psi\rangle \text{ in } \mathcal{R}. \tag{2.1.6}$$

An elementary theorem tells us that for A Hermitian, α is real, and eigenvectors with different αs are orthogonal.

[1] We shall often use the Dirac bra-ket notation: instead of $\langle\Psi_1|\Psi_2\rangle$, we may write $\langle 1|2\rangle$.

(iii) If a system is in a state represented by a ray $\mathcal{R}$, and an experiment is done to test whether it is in any one of the different states represented by mutually orthogonal rays $\mathcal{R}_1, \mathcal{R}_2, \ldots$ (for instance, by measuring one or more observables) then the probability of finding it in the state represented by $\mathcal{R}_n$ is

$$P(\mathcal{R} \to \mathcal{R}_n) = \big|\langle\Psi|\Psi_n\rangle\big|^2, \tag{2.1.7}$$

where $|\Psi\rangle$ and $|\Psi_n\rangle$ are any vectors belonging to rays $\mathcal{R}$ and $\mathcal{R}_n$, respectively. (A pair of rays is said to be orthogonal if the state-vectors from the two rays have vanishing scalar products.) Another elementary theorem gives a total probability unity:

$$\sum_n P(\mathcal{R} \to \mathcal{R}_n) = 1 \tag{2.1.8}$$

if the state-vectors $|\Psi_n\rangle$ form a complete set.

2.2 Symmetries

A symmetry transformation is a change in our point of view that does not change the results of possible experiments. If an observer O sees a system in a state represented by a ray $\mathcal{R}$ or $\mathcal{R}_1$ or $\mathcal{R}_2, \ldots$, then an equivalent observer O' who looks at the *same* system will observe it in a different state, represented by a ray $\mathcal{R}'$ or $\mathcal{R}'_1$ or $\mathcal{R}'_2, \ldots$, respectively, but the two observers must find the same probabilities

$$P(\mathcal{R} \to \mathcal{R}_n) = P(\mathcal{R}' \to \mathcal{R}'_n). \tag{2.2.1}$$

(This is only a necessary condition for a ray transformation to be a symmetry; further conditions are discussed in the next chapter.) An important theorem proved by Wigner [2] in the early 1930s tells us that for any such transformation $\mathcal{R} \to \mathcal{R}'$ of rays we may define an operator U on Hilbert space, such that if $|\Psi\rangle$ is in ray $\mathcal{R}$ then $U|\Psi\rangle$ is in the ray $\mathcal{R}'$, with U either *unitary* and *linear*

$$\langle U\Phi|U\Psi\rangle = \langle\Phi|\Psi\rangle, \tag{2.2.2}$$

$$U\big(\xi|\Phi\rangle + \eta|\Psi\rangle\big) = \xi U|\Phi\rangle + \eta U|\Psi\rangle \tag{2.2.3}$$

or else *antiunitary* and *antilinear*

$$\langle U\Phi|U\Psi\rangle = \langle\Phi|\Psi\rangle^*, \tag{2.2.4}$$

$$U\big(\xi|\Phi\rangle + \eta|\Psi\rangle\big) = \xi^* U|\Phi\rangle + \eta^* U|\Psi\rangle. \tag{2.2.5}$$

Wigner's proof omits some steps. A more complete proof is given at the end of this chapter in Appendix A.

As already mentioned, the adjoint of a linear operator L is defined by

$$\langle\Phi|L^\dagger|\Psi\rangle \equiv \langle L\Phi|\Psi\rangle. \tag{2.2.6}$$

This condition cannot be satisfied for an antilinear operator, because in this case the right-hand side of Eq. (2.2.6) would be linear in $|\Phi\rangle$, while the left-hand side is antilinear in $|\Phi\rangle$. Instead, the adjoint of an antilinear operator A is defined by

$$\langle\Phi|A^\dagger|\Psi\rangle \equiv \langle A\Phi|\Psi\rangle^* = \langle\Psi|A|\Phi\rangle. \tag{2.2.7}$$

With this definition, the conditions for unitarity or antiunitarity both take the form

$$U^\dagger = U^{-1}. \tag{2.2.8}$$

There is always a trivial symmetry transformation $\mathcal{R} \to \mathcal{R}$, represented by the identity operator $U = \mathbf{1}$. This operator is, of course, unitary and linear. Continuity then demands that any symmetry (like a rotation or translation or Lorentz transformation) that can be made trivial by a continuous change of some parameters (like angles or distances or velocities) must be represented by a linear unitary operator U rather than one that is antilinear and antiunitary. (Symmetries represented by antiunitary antilinear operators are less prominent in physics; they all involve a reversal in the direction of time's flow. See Section 2.6.)

In particular, a symmetry transformation that is infinitesimally close to being trivial can be represented by a linear unitary operator that is infinitesimally close to the identity:

$$U = \mathbf{1} + i\epsilon t \tag{2.2.9}$$

with ϵ a real infinitesimal. For this to be unitary and linear, t must be Hermitian and linear, so it is a candidate for an observable. Indeed, most (and perhaps all) of the observables of physics, such as angular momentum or momentum, arise in this way from symmetry transformations.

The set of symmetry transformations has certain properties that define it as a *group*. If T_1 is a transformation that takes rays $\mathcal{R}_n$ into $\mathcal{R}'_n$, and T_2 is another transformation that takes $\mathcal{R}'_n$ into $\mathcal{R}''_n$, then the result of performing both transformations is another symmetry transformation, which we write T_2T_1, that takes $\mathcal{R}_n$ into $\mathcal{R}''_n$. Also, a symmetry transformation T which takes rays $\mathcal{R}_n$ into $\mathcal{R}'_n$ has an inverse, written T^{-1}, which takes $\mathcal{R}'_n$ into $\mathcal{R}_n$, and there is an identity transformation, $T = \mathbf{1}$, which leaves rays unchanged.

The unitary or antiunitary operators $U(T)$ corresponding to these symmetry transformations have properties that mirror this group structure, but with a complication due to the fact that, unlike the symmetry transformations themselves, the operators $U(T)$ act on vectors in the Hilbert space, rather than on rays. If T_1 takes $\mathcal{R}_n$ into $\mathcal{R}'_n$, then acting on a vector $|\Psi_n\rangle$ in the ray $\mathcal{R}_n$, $U(T_1)$ must yield a vector $U(T_1)|\Psi_n\rangle$ in the ray $\mathcal{R}'_n$, and if T_2 takes this ray into $\mathcal{R}''_n$, then acting on $U(T_1)|\Psi_n\rangle$ it must yield a vector $U(T_2)U(T_1)|\Psi_n\rangle$ in the ray $\mathcal{R}''_n$. But $U(T_2T_1)|\Psi_n\rangle$ is also in this ray, so these vectors can differ only by a phase $\phi_n(T_2, T_1)$

$$U(T_2)U(T_1)|\Psi_n\rangle = e^{i\phi_n(T_2,T_1)}U(T_2T_1)|\Psi_n\rangle. \tag{2.2.10}$$

Furthermore, with one significant exception, the linearity (or antilinearity) of $U(T)$ tells us that these phases are independent of the state $|\Psi_n\rangle$. Here is the proof. Consider any two different vectors $|\Psi_A\rangle, |\Psi_B\rangle$, which are not proportional to each other. Then, applying Eq. (2.2.10) to the state $|\Psi_{AB}\rangle = |\Psi_A\rangle + |\Psi_B\rangle$, we have

$$\begin{aligned} e^{i\phi_{AB}}U(T_2T_1)\big(|\Psi_A\rangle + |\Psi_B\rangle\big) &= U(T_2)U(T_1)\big(|\Psi_A\rangle + |\Psi_B\rangle\big) \\ &= U(T_2)U(T_1)|\Psi_A\rangle + U(T_2)U(T_1)|\Psi_B\rangle \\ &= e^{i\phi_A}U(T_2T_1)|\Psi_A\rangle + e^{i\phi_B}U(T_2T_1)|\Psi_B\rangle. \end{aligned} \tag{2.2.11}$$

Any unitary or antiunitary operator has an inverse (its adjoint) which is also unitary or antiunitary. Multiplying (2.2.11) on the left with $U^{-1}(T_2T_1)$, we have then

$$e^{\pm i\phi_{AB}}\big(|\Psi_A\rangle + |\Psi_B\rangle\big) = e^{\pm i\phi_A}|\Psi_A\rangle + e^{\pm i\phi_B}|\Psi_B\rangle, \tag{2.2.12}$$

the upper and lower signs referring to $U(T_2T_1)$ unitary or antiunitary, respectively. Since $|\Psi_A\rangle$ and $|\Psi_B\rangle$ are linearly independent, this is only possible if

$$e^{i\phi_{AB}} = e^{i\phi_A} = e^{i\phi_B}. \tag{2.2.13}$$

So as promised, the phase in Eq. (2.2.10) is independent of the state-vector $|\Psi_n\rangle$, and therefore this can be written as an operator relation

$$U(T_2)U(T_1) = e^{i\phi(T_2,T_1)}U(T_2T_1). \tag{2.2.14}$$

For $\phi = 0$, this would say that the $U(T)$ furnish a representation of the group of symmetry transformations. For general phases $\phi(T_2, T_1)$, we have what is called a projective representation, or a representation 'up to a phase'. The structure of the Lie group cannot by itself tell us whether physical state-vectors furnish an ordinary or a projective representation, but as we shall see, it can tell us whether the group has any intrinsically projective representations at all.

The exception to the argument that led to Eq. (2.2.14) is that it may not be possible to prepare the system in a state represented by $|\Psi_A\rangle + |\Psi_B\rangle$. For instance, it is widely believed to be impossible to prepare a system in a superposition of two states whose total angular momenta are integers and half-integers, respectively. In such cases, we say that there is a 'superselection rule' between different classes of states [3], and the phases $\phi(T_2, T_1)$ may depend on which of these classes of states the operators $U(T_2)U(T_1)$ and $U(T_2T_1)$ act upon. We will have more to say about these phases and projective representations in Section 2.7. As we shall see there, any symmetry group with projective representations can always be enlarged (without otherwise changing its physical implications) in such a way that its representations can all be defined as non-projective, with $\phi = 0$. Until Section 2.7, we will just assume that this has been done, and take $\phi = 0$ in Eq. (2.2.14).

There is a kind of group, known as a *connected Lie group*, of special importance in physics. These are groups of transformations $T(\theta)$ that are described by a finite set of real continuous parameters, say θ^a, with each element of the group connected to the identity by a path within the group. The group multiplication law then takes the form

$$T(\bar\theta)T(\theta) = T\big(f(\bar\theta,\theta)\big) \tag{2.2.15}$$

with $f^a(\bar\theta,\theta)$ a function of the $\bar\theta$s and θs. Taking $\theta^a = 0$ as the coordinates of the identity, we must have

$$f^a(\theta,0) = f^a(0,\theta) = \theta^a. \tag{2.2.16}$$

As already mentioned, the transformations of such continuous groups must be represented on the physical Hilbert space by unitary (rather than antiunitary) operators $U(T(\theta))$. For a Lie group, these operators can be represented in at least a finite neighborhood of the identity by a power series

$$U\big(T(\theta)\big) = \mathbf{1} + i\theta^a t_a + \frac{1}{2}\theta^b\theta^c t_{bc} + \cdots\,, \tag{2.2.17}$$

where t_a, $t_{bc} = t_{cb}$, etc. are Hermitian operators independent of the θs. Suppose that the $U(T(\theta))$ form an ordinary (i.e., not projective) representation of this group of transformations, i.e.,

$$U\big(T(\bar\theta)\big)U\big(T(\theta)\big) = U\big(T(f(\bar\theta,\theta))\big). \tag{2.2.18}$$

Let us see what this condition looks like when expanded in powers of θ^a and $\bar\theta^a$. According to Eq. (2.2.16), the expansion of $f^a(\bar\theta,\theta)$ to second order must take the form

$$f^a(\bar\theta,\theta) = \theta^a + \bar\theta^a + f^a{}_{bc}\bar\theta^b\theta^c + \cdots \tag{2.2.19}$$

with real coefficients $f^a{}_{bc}$. (The presence of any terms of order θ^2 or $\bar\theta^2$ would violate Eq. (2.2.16).) Then Eq. (2.2.18) reads

$$\begin{aligned}
&\left[\mathbf{1} + i\bar\theta^a t_a + \frac{1}{2}\bar\theta^b\bar\theta^c t_{bc} + \cdots\right]\left[\mathbf{1} + i\theta^a t_a + \frac{1}{2}\theta^b\theta^c t_{bc} + \cdots\right] \\
&\quad = \mathbf{1} + i\big(\theta^a + \bar\theta^a + f^a{}_{bc}\bar\theta^b\theta^c + \cdots\big)t_a \\
&\qquad + \frac{1}{2}\big(\theta^b + \bar\theta^b + \cdots\big)\big(\theta^c + \bar\theta^c + \cdots\big)t_{bc} + \cdots\,.
\end{aligned} \tag{2.2.20}$$

The terms of order 1, θ, $\bar\theta$, θ^2, and $\bar\theta^2$ automatically match on both sides of Eq. (2.2.20), but from the $\bar\theta\theta$ terms we obtain a non-trivial condition

$$t_{bc} = -t_b t_c - i f^a{}_{bc} t_a. \tag{2.2.21}$$

This shows that if we are given the structure of the group, i.e., the function $f(\theta,\bar\theta)$, and hence its quadratic coefficient $f^a{}_{bc}$, we can calculate the second-order terms in $U(T(\theta))$ from the generators t_a appearing in the first-order terms. However, there is a consistency condition: the operator t_{bc} must be *symmetric* in b and c (because it is the second derivative of $U(T(\theta))$ with respect to θ^b and θ^c) so Eq. (2.2.21) requires that

$$[t_b, t_c] = iC^a{}_{bc} t_a, \tag{2.2.22}$$

where $C^a{}_{bc}$ are a set of real constants known as *structure constants*

$$C^a{}_{bc} \equiv -f^a{}_{bc} + f^a{}_{cb}. \tag{2.2.23}$$

Such a set of commutation relations is known as a *Lie algebra.* In Section 2.7 we will prove in effect that the commutation relation (2.2.22) is the single condition needed to ensure that this process can be continued: the complete power series for $U(T(\theta))$ may be calculated from an infinite sequence of relations like Eq. (2.2.21), provided we know the first-order terms, the generators t_a. This does not necessarily mean that the operators $U(T(\theta))$ are uniquely determined for all θ^a if we know the t_a, but it does mean that the $U(T(\theta))$ are uniquely determined in at least a finite neighborhood of the coordinates $\theta^a = 0$ of the identity, in such a way that Eq. (2.2.15) is satisfied if θ, $\bar{\theta}$, and $f(\theta, \bar{\theta})$ are in this neighborhood. The extension to all θ^a is discussed in Section 2.7.

There is a special case of some importance, that we will encounter again and again. Suppose that the function $f(\theta, \bar{\theta})$ (perhaps just for some subset of the coordinates θ^a) is simply additive

$$f^a(\theta, \bar{\theta}) = \theta^a + \bar{\theta}^a. \tag{2.2.24}$$

This is the case for instance for translations in spacetime, or for rotations about any one fixed axis (though *not* for both together). Then the coefficients ${f^a}_{bc}$ in Eq. (2.2.19) vanish, and so do the structure constants (2.2.23). The generators then all commute

$$[t_b, t_c] = 0. \tag{2.2.25}$$

Such a group is called *Abelian*. In this case, it is easy to calculate $U(T(\theta))$ for all θ^a. From Eqs. (2.2.18) and (2.2.24), we have for any integer N

$$U\big(T(\theta)\big) = \left[U\left(T\left(\frac{\theta}{N}\right)\right)\right]^N.$$

Letting $N \to \infty$, and keeping only the first-order term in $U(T(\theta/N))$, we have then

$$U\big(T(\theta)\big) = \lim_{N \to \infty} \left[\mathbf{1} + \frac{i}{N}\theta^a t_a\right]^N$$

and hence

$$U\big(T(\theta)\big) = \exp(i t_a \theta^a). \tag{2.2.26}$$

2.3 Quantum Lorentz Transformations

Einstein's principle of relativity states the equivalence of certain 'inertial' frames of reference. It is distinguished from the Galilean principle of relativity, obeyed by Newtonian mechanics, by the transformation connecting coordinate systems in different inertial frames. If x^μ are the coordinates in one inertial frame (with $x^0 = t$ a time coordinate and x^1, x^2, x^3 Cartesian space coordinates, the speed of light being set equal to unity) then in any other inertial frame, the coordinates x'^μ must satisfy

$$\eta_{\mu\nu}\, dx'^\mu dx'^\nu = \eta_{\mu\nu}\, dx^\mu dx^\nu \tag{2.3.1}$$

or equivalently

$$\eta_{\mu\nu} \frac{\partial x'^\mu}{\partial x^\rho} \frac{\partial x'^\nu}{\partial x^\sigma} = \eta_{\rho\sigma}. \tag{2.3.2}$$

Here $\eta_{\mu\nu}$ is the diagonal matrix, with elements

$$\eta_{00} = -1, \qquad \eta_{11} = \eta_{22} = \eta_{33} = +1, \tag{2.3.3}$$

and the summation convention is in force: we sum over any index like μ and ν in Eq. (2.3.2), which appears twice in the same term, once upstairs and once downstairs. These transformations have the special property that the speed of light is the same (in our units, equal to unity) in all inertial frames;[2] a light wave travelling at unit speed satisfies $|d\mathbf{x}/dt| = 1$, or in other words $\eta_{\mu\nu} dx^\mu dx^\nu = d\mathbf{x}^2 - dt^2 = 0$, from which it follows that also $\eta_{\mu\nu} dx'^\mu dx'^\nu = 0$, and hence $|d\mathbf{x}'/dt'| = 1$.

Any coordinate transformation $x^\mu \to x'^\mu$ that satisfies Eq. (2.3.2) is *linear* [3a]

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu + a^\mu \tag{2.3.4}$$

with a^μ arbitrary constants, and $\Lambda^\mu{}_\nu$ a constant matrix satisfying the conditions

$$\eta_{\mu\nu} \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma = \eta_{\rho\sigma}. \tag{2.3.5}$$

For some purposes, it is useful to write the Lorentz transformation condition in a different way. The matrix $\eta_{\mu\nu}$ has an inverse, written $\eta^{\mu\nu}$, which happens to have the same components: it is diagonal, with $\eta^{00} = -1$, $\eta^{11} = \eta^{22} = \eta^{33} = +1$. Multiplying Eq. (2.3.5) with $\eta^{\sigma\tau}\Lambda^\kappa{}_\tau$ and inserting parentheses judiciously, we have

$$\eta_{\mu\nu}\Lambda^\mu{}_\rho \big(\Lambda^\nu{}_\sigma \Lambda^\kappa{}_\tau \eta^{\sigma\tau}\big) = \Lambda^\kappa{}_\rho = \eta_{\mu\nu}\eta^{\nu\kappa}\Lambda^\mu{}_\rho.$$

Multiplying with the inverse of the matrix $\eta_{\mu\nu}\Lambda^\mu{}_\rho$ then gives

$$\Lambda^\nu{}_\sigma \Lambda^\kappa{}_\tau \eta^{\sigma\tau} = \eta^{\nu\kappa}. \tag{2.3.6}$$

These transformations form a group. If we first perform a Lorentz transformation (2.3.4), and then a second Lorentz transformation $x'^\mu \to x''^\mu$, with

$$x''^\mu = \bar\Lambda^\mu{}_\rho x'^\rho + \bar a^\mu = \bar\Lambda^\mu{}_\rho \big(\Lambda^\rho{}_\nu x^\nu + a^\rho\big) + \bar a^\mu$$

then the effect is the same as the Lorentz transformation $x^\mu \to x''^\mu$, with

$$x''^\mu = \big(\bar\Lambda^\mu{}_\rho \Lambda^\rho{}_\nu\big) x^\nu + \big(\bar\Lambda^\mu{}_\rho a^\rho + \bar a^\mu\big). \tag{2.3.7}$$

(Note that if $\Lambda^\mu{}_\nu$ and $\bar\Lambda^\mu{}_\nu$ both satisfy Eq. (2.3.5), then so does $\bar\Lambda^\mu{}_\rho \Lambda^\rho{}_\nu$, so this is a Lorentz transformation. The bar is used here just to distinguish one Lorentz transformation from the other.) The transformations $T(\Lambda, a)$ induced on physical states therefore satisfy the composition rule

$$T(\bar\Lambda, \bar a) T(\Lambda, a) = T(\bar\Lambda\Lambda, \bar\Lambda a + \bar a). \tag{2.3.8}$$

Taking the determinant of Eq. (2.3.5) gives

$$(\mathrm{Det}\,\Lambda)^2 = 1 \tag{2.3.9}$$

so $\Lambda^\mu{}_\nu$ has an inverse, $(\Lambda^{-1})^\nu{}_\mu$, which we see from Eq. (2.3.5) takes the form

$$(\Lambda^{-1})^\nu{}_\mu = \Lambda_\mu{}^\nu \equiv \eta_{\mu\rho}\eta^{\nu\sigma}\Lambda^\rho{}_\sigma. \tag{2.3.10}$$

[2]There is a larger class of coordinate transformations, known as *conformal transformations*, for which $\eta_{\mu\nu} dx'^\mu dx'^\nu$ is proportional though generally not equal to $\eta_{\mu\nu} dx^\mu dx^\nu$, and which therefore also leave the speed of light invariant. Conformal invariance in two dimensions has proved enormously important in string theory and statistical mechanics, but the physical relevance of these conformal transformations in four spacetime dimensions is not yet clear.

The inverse of the transformation $T(\Lambda, a)$ is seen from Eq. (2.3.8) to be $T(\Lambda^{-1}, -\Lambda^{-1}a)$, and, of course, the identity transformation is $T(\mathbf{1}, 0)$.

In accordance with the discussion in the previous section, the transformations $T(\Lambda, a)$ induce a unitary linear transformation on vectors in the physical Hilbert space

$$|\Psi\rangle \longrightarrow U(\Lambda, a)|\Psi\rangle.$$

The operators U satisfy a composition rule

$$U(\bar{\Lambda}, \bar{a})U(\Lambda, a) = U(\bar{\Lambda}\Lambda, \bar{\Lambda}a + \bar{a}). \tag{2.3.11}$$

(As already mentioned, to avoid the appearance of a phase factor on the right-hand side of Eq. (2.3.11), it is, in general, necessary to enlarge the Lorentz group. The appropriate enlargement is described in Section 2.7.)

The whole group of transformations $T(\Lambda, a)$ is properly known as the *inhomogeneous Lorentz group*, or *Poincaré group*. It has a number of important subgroups. First, those transformations with $a^\mu = 0$ obviously form a subgroup, with

$$T(\bar{\Lambda}, 0)T(\Lambda, 0) = T(\bar{\Lambda}\Lambda, 0), \tag{2.3.12}$$

known as the *homogeneous Lorentz group*. Also, we note from Eq. (2.3.9) that either $\operatorname{Det}\Lambda = +1$ or $\operatorname{Det}\Lambda = -1$; those transformations with $\operatorname{Det}\Lambda = +1$ obviously form a subgroup of either the homogeneous or the inhomogeneous Lorentz groups. Further, from the 00-components of Eqs. (2.3.5) and (2.3.6), we have

$$(\Lambda^0{}_0)^2 = 1 + \Lambda^i{}_0\Lambda^i{}_0 = 1 + \Lambda^0{}_i\Lambda^0{}_i, \tag{2.3.13}$$

with i summed over the values 1, 2, and 3. We see that either $\Lambda^0{}_0 \geq +1$ or $\Lambda^0{}_0 \leq -1$. Those transformations with $\Lambda^0{}_0 \geq +1$ form a subgroup. Note that if $\Lambda^\mu{}_\nu$ and $\bar{\Lambda}^\mu{}_\nu$ are two such Λs, then

$$(\bar{\Lambda}\Lambda)^0{}_0 = \bar{\Lambda}^0{}_0\Lambda^0{}_0 + \bar{\Lambda}^0{}_1\Lambda^1{}_0 + \bar{\Lambda}^0{}_2\Lambda^2{}_0 + \bar{\Lambda}^0{}_3\Lambda^3{}_0.$$

But Eq. (2.3.13) shows that the three-vector $(\Lambda^1{}_0, \Lambda^2{}_0, \Lambda^3{}_0)$ has length $\sqrt{(\Lambda^0{}_0)^2 - 1}$, and similarly the three-vector
$(\bar{\Lambda}^0{}_1, \bar{\Lambda}^0{}_2, \bar{\Lambda}^0{}_3)$ has length $\sqrt{(\bar{\Lambda}^0{}_0)^2 - 1}$, so the scalar product of these two three-vectors is bounded by

$$\left|\bar{\Lambda}^0{}_1\Lambda^1{}_0 + \bar{\Lambda}^0{}_2\Lambda^2{}_0 + \bar{\Lambda}^0{}_3\Lambda^3{}_0\right| \leq \sqrt{(\Lambda^0{}_0)^2 - 1}\sqrt{(\bar{\Lambda}^0{}_0)^2 - 1}, \tag{2.3.14}$$

and so

$$(\bar{\Lambda}\Lambda)^0{}_0 \geq \bar{\Lambda}^0{}_0\Lambda^0{}_0 - \sqrt{(\Lambda^0{}_0)^2 - 1}\sqrt{(\bar{\Lambda}^0{}_0)^2 - 1} \geq 1.$$

The subgroup of Lorentz transformations with $\operatorname{Det}\Lambda = +1$ and $\Lambda^0{}_0 \geq +1$ is known as the *proper orthochronous Lorentz group*. Since it is not possible by a continuous change of parameters to jump from $\operatorname{Det}\Lambda = +1$ to $\operatorname{Det}\Lambda = -1$, or from $\Lambda^0{}_0 \geq +1$ to $\Lambda^0{}_0 \leq -1$, any Lorentz transformation that can be obtained from the identity by a continuous change of parameters must have $\operatorname{Det}\Lambda$ and $\Lambda^0{}_0$ of the same sign as for the identity, and hence must belong to the proper orthochronous Lorentz group.

Any Lorentz transformation is either proper and orthochronous, or may be written as the product of an element of the proper orthochronous Lorentz group with one of the discrete transformations $\mathcal{P}$ or $\mathcal{T}$ or $\mathcal{PT}$, where $\mathcal{P}$ is the space inversion, whose non-zero elements are

$$\mathcal{P}^0{}_0 = 1, \qquad \mathcal{P}^1{}_1 = \mathcal{P}^2{}_2 = \mathcal{P}^3{}_3 = -1, \tag{2.3.15}$$

and $\mathcal{T}$ is the time-reversal matrix, whose non-zero elements are

$$\mathcal{T}^0{}_0 = -1, \qquad \mathcal{T}^1{}_1 = \mathcal{T}^2{}_2 = \mathcal{T}^3{}_3 = 1. \tag{2.3.16}$$

Thus the study of the whole Lorentz group reduces to the study of its proper orthochronous subgroup, plus space inversion and time-reversal. We will consider space inversion and time-reversal separately in Section 2.6. Until then, we will deal only with the homogeneous or inhomogeneous proper orthochronous Lorentz group.

2.4 The Poincaré Algebra

As we saw in Section 2.2, much of the information about any Lie symmetry group is contained in properties of the group elements near the identity. For the inhomogeneous Lorentz group, the identity is the transformation $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu$, $a^\mu = 0$, so we want to study those transformations with

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu\,, \qquad a^\mu = \epsilon^\mu\,, \tag{2.4.1}$$

both $\omega^\mu{}_\nu$ and ϵ^μ being taken infinitesimal. The Lorentz condition (2.3.5)

reads here

$$\begin{aligned}\eta_{\rho\sigma} &= \eta_{\mu\nu}(\delta^\mu_\rho + \omega^\mu{}_\rho)(\delta^\nu_\sigma + \omega^\nu{}_\sigma)\\ &= \eta_{\rho\sigma} + \omega_{\sigma\rho} + \omega_{\rho\sigma} + O(\omega^2)\,.\end{aligned}$$

We are here using the convention, to be used throughout this book, that indices may be lowered or raised by contraction with $\eta_{\mu\nu}$ or $\eta^{\mu\nu}$

$$\omega_{\sigma\rho} = \eta_{\mu\sigma}\omega^\mu{}_\rho, \qquad \omega^\mu{}_\rho = \eta^{\mu\sigma}\omega_{\sigma\rho}\,.$$

Keeping only the terms of first order in ω in the Lorentz condition (2.3.5), we see that this condition now reduces to the antisymmetry of $\omega_{\mu\nu}$

$$\omega_{\mu\nu} = -\omega_{\nu\mu}\,. \tag{2.4.2}$$

An antisymmetric second-rank tensor in four dimensions has $(4\times 3)/2 = 6$ independent components, so including the four components of ϵ^μ, an inhomogeneous Lorentz transformation is described by $6 + 4 = 10$ parameters.

Since $U(\mathbf{1}, 0)$ carries any ray into itself, it must be proportional to the unit operator,[3] and by a choice of phase may be made equal to it. For an infinitesimal Lorentz transformation (2.4.1), $U(\mathbf{1} + \omega, \epsilon)$ must then equal $\mathbf{1}$ plus terms linear in $\omega_{\rho\sigma}$ and ϵ_ρ. We write

[3] In the absence of superselection rules, the possibility that the proportionality factor may depend on the state on which $U(\mathbf{1}, 0)$ acts can be ruled out by the same reasoning that we used in Section 2.2 to rule out the possibility that the phases in projective representations of symmetry groups may depend on the states on which the symmetries act. Where superselection rules apply, it may be necessary to redefine $U(\mathbf{1}, 0)$ by phase factors that depend on the sector on which it acts.

this as

$$U(\mathbf{1}+\omega,\epsilon)=\mathbf{1}+\frac{i}{2}\omega_{\rho\sigma}J^{\rho\sigma}-i\epsilon_\rho P^\rho+\cdots\,. \tag{2.4.3}$$

Here $J^{\rho\sigma}$ and P^ρ are ω- and ϵ-independent operators, and the dots denote terms of higher order in ω and/or ϵ. In order for $U(\mathbf{1}+\omega,\epsilon)$ to be unitary, the operators $J^{\rho\sigma}$ and P^ρ must be Hermitian

$$J^{\rho\sigma\dagger}=J^{\rho\sigma}\,,\qquad P^{\rho\dagger}=P^\rho\,. \tag{2.4.4}$$

Since $\omega_{\rho\sigma}$ is antisymmetric, we can take its coefficient $J^{\rho\sigma}$ to be antisymmetric also

$$J^{\rho\sigma}=-J^{\sigma\rho}\,. \tag{2.4.5}$$

As we shall see, P^1, P^2, and P^3 are the components of the momentum operators, J^{23}, J^{31}, and J^{12} are the components of the angular momentum vector, and P^0 is the energy operator, or Hamiltonian.[4]

We now examine the Lorentz transformation properties of $J^{\rho\sigma}$ and P^ρ. We consider the product

$$U(\Lambda,a)\,U(\mathbf{1}+\omega,\epsilon)\,U^{-1}(\Lambda,a)\,,$$

where $\Lambda^\mu{}_\nu$ and a^μ are here the parameters of a new transformation, unrelated to ω and ϵ. According to Eq. (2.3.11), the product $U(\Lambda^{-1},-\Lambda^{-1}a)U(\Lambda,a)$ equals $U(\mathbf{1},0)$, so $U(\Lambda^{-1},-\Lambda^{-1}a)$ is the inverse of $U(\Lambda,a)$. It follows then from (2.3.11) that

$$\begin{aligned}&U(\Lambda,a)U(\mathbf{1}+\omega,\epsilon)U^{-1}(\Lambda,a)\\&\quad=U\big(\Lambda(\mathbf{1}+\omega)\Lambda^{-1},\Lambda\epsilon-\Lambda\omega\Lambda^{-1}a\big)\,.\end{aligned} \tag{2.4.6}$$

To first order in ω and ϵ, we have then

$$\begin{aligned}&U(\Lambda,a)\left[\frac{1}{2}\omega_{\rho\sigma}J^{\rho\sigma}-\epsilon_\rho P^\rho\right]U^{-1}(\Lambda,a)\\&\quad=\frac{1}{2}(\Lambda\omega\Lambda^{-1})_{\mu\nu}J^{\mu\nu}-(\Lambda\epsilon-\Lambda\omega\Lambda^{-1}a)_\mu P^\mu\,.\end{aligned} \tag{2.4.7}$$

Equating coefficients of $\omega_{\rho\sigma}$ and ϵ_ρ on both sides of this equation (and using (2.3.10)), we find

$$U(\Lambda,a)J^{\rho\sigma}U^{-1}(\Lambda,a)=\Lambda_\mu{}^\rho\Lambda_\nu{}^\sigma\left(J^{\mu\nu}-a^\mu P^\nu+a^\nu P^\mu\right), \tag{2.4.8}$$

$$U(\Lambda,a)P^\rho U^{-1}(\Lambda,a)=\Lambda_\mu{}^\rho P^\mu\,. \tag{2.4.9}$$

For homogeneous Lorentz transformations (with $a^\mu=0$), these transformation rules simply say that $J^{\mu\nu}$ is a tensor and P^μ is a vector. For pure translations (with $\Lambda^\mu{}_\nu=\delta^\mu{}_\nu$), they tell us that P^ρ is translation-invariant, but $J^{\rho\sigma}$ is not. In particular, the change of the space-space components of $J^{\rho\sigma}$ under a spatial translation is just the usual change of the angular momentum under a change of the origin relative to which the angular momentum is calculated.

[4] We will see that this identification of the angular-momentum generators is forced on us by the commutation relations of the $J^{\mu\nu}$. On the other hand, the commutation relations do not allow us to distinguish between P^μ and $-P^\mu$, so the sign for the $\epsilon_\rho P^\rho$ term in (2.4.3) is a matter of convention. The consistency of the choice in (2.4.3) with the usual definition of the Hamiltonian P^0 is shown in Section 3.1.

Next, let's apply rules (2.4.8), (2.4.9) to a transformation that is itself infinitesimal, i.e., $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega'^\mu{}_\nu$ and $a^\mu = \epsilon'^\mu$, with infinitesimals $\omega'^\mu{}_\nu$ and ϵ'^μ unrelated to the previous ω and ϵ. Using Eq. (2.4.3), and keeping only terms of first order in $\omega'^\mu{}_\nu$ and ϵ'^μ, Eqs. (2.4.8) and (2.4.9) now become

$$i\left[\frac{1}{2}\omega'_{\mu\nu}J^{\mu\nu} - \epsilon'_\mu P^\mu, J^{\rho\sigma}\right] = \omega'^\rho{}_\mu J^{\mu\sigma} + \omega'^\sigma{}_\nu J^{\rho\nu} - \epsilon'^\rho P^\sigma + \epsilon'^\sigma P^\rho\,, \tag{2.4.10}$$

$$i\left[\frac{1}{2}\omega'_{\mu\nu}J^{\mu\nu} - \epsilon'_\mu P^\mu, P^\rho\right] = \omega'^\rho{}_\mu P^\mu\,. \tag{2.4.11}$$

Equating coefficients of $\omega'_{\mu\nu}$ and ϵ'_μ on both sides of these equations, we find the commutation rules

$$i[J^{\mu\nu}, J^{\rho\sigma}] = \eta^{\nu\rho}J^{\mu\sigma} - \eta^{\mu\rho}J^{\nu\sigma} - \eta^{\nu\sigma}J^{\mu\rho} + \eta^{\mu\sigma}J^{\nu\rho}\,, \tag{2.4.12}$$

$$i[P^\mu, J^{\rho\sigma}] = \eta^{\mu\rho}P^\sigma - \eta^{\mu\sigma}P^\rho\,, \tag{2.4.13}$$

$$[P^\mu, P^\rho] = 0\,. \tag{2.4.14}$$

This is the Lie algebra of the Poincaré group.

In quantum mechanics a special role is played by those operators that are *conserved*, i.e., that commute with the energy operator $H = P^0$. Inspection of Eqs. (2.4.13) and (2.4.14) shows that these are the momentum three-vector

$$\mathbf{P} = \{P^1, P^2, P^3\} \tag{2.4.15}$$

and the angular-momentum three-vector

$$\mathbf{J} = \{J^{23}, J^{31}, J^{12}\} \tag{2.4.16}$$

and, of course, the energy P^0 itself. The remaining generators form what is called the 'boost' three-vector

$$\mathbf{K} = \{J^{10}, J^{20}, J^{30}\}\,. \tag{2.4.17}$$

These are *not* conserved, which is why we do not use the eigenvalues of $\mathbf{K}$ to label physical states. In a three-dimensional notation, the commutation relations (2.4.12), (2.4.13), (2.4.14) may be written

$$[J_i, J_j] = i\epsilon_{ijk}J_k\,, \tag{2.4.18}$$

$$[J_i, K_j] = i\epsilon_{ijk}K_k\,, \tag{2.4.19}$$

$$[K_i, K_j] = -i\epsilon_{ijk}J_k\,, \tag{2.4.20}$$

$$[J_i, P_j] = i\epsilon_{ijk}P_k\,, \tag{2.4.21}$$

$$[K_i, P_j] = iH\delta_{ij}\,, \tag{2.4.22}$$

$$[J_i, H] = [P_i, H] = [H, H] = 0\,, \tag{2.4.23}$$

$$[K_i, H] = iP_i\,, \tag{2.4.24}$$

where i, j, k, etc. run over the values 1, 2, and 3, and ϵ_{ijk} is the totally antisymmetric quantity with $\epsilon_{123} = +1$. The commutation relation (2.4.18) will be recognized as that of the angular-momentum operator.

The pure translations $T(\mathbf{1}, a)$ form a subgroup of the inhomogeneous Lorentz group with a group multiplication rule given by (2.3.7) as

$$T(\mathbf{1}, \bar{a})T(\mathbf{1}, a) = T(\mathbf{1}, \bar{a} + a)\,. \tag{2.4.25}$$

This is additive in the same sense as (2.2.24), so by using (2.4.3) and repeating the same arguments that led to (2.2.26), we find that finite translations are represented on the physical Hilbert space by

$$U(\mathbf{1}, a) = \exp(-iP^\mu a_\mu)\,. \tag{2.4.26}$$

In exactly the same way, we can show that a rotation $R_{\boldsymbol{\theta}}$ by an angle $|\boldsymbol{\theta}|$ around the direction of $\boldsymbol{\theta}$ is represented on the physical Hilbert space by

$$U(R_{\boldsymbol{\theta}}, 0) = \exp(i\mathbf{J} \cdot \boldsymbol{\theta})\,. \tag{2.4.27}$$

It is interesting to compare the Poincaré algebra with the Lie algebra of the symmetry group of Newtonian mechanics, the Galilean group. We could derive this algebra by starting with the transformation rules of the Galilean group and then following the same procedure that was used here to derive the Poincaré algebra. However, since we already have Eqs. (2.4.18)–(2.4.24), it is easier to obtain the Galilean algebra as a low-velocity limit of the Poincaré algebra, by what is known as an *Inönü–Wigner contraction* [4,5]. For a system of particles of typical mass m and typical velocity v, the momentum and the angular-momentum operators are expected to be of order $J \sim 1$, $P \sim mv$. On the other hand, the energy operator is $H = M + W$ with a total mass M and non-mass energy W (kinetic plus potential) of order $M \sim m$, $W \sim mv^2$. Inspection of Eqs. (2.4.18)–(2.4.24) shows that these commutation relations have a limit for $v \ll 1$ of the form

$$\begin{gathered}
[J_i, J_j] = i\epsilon_{ijk}J_k\,, \qquad [J_i, K_j] = i\epsilon_{ijk}K_k\,, \qquad [K_i, K_j] = 0\,, \\
[J_i, P_j] = i\epsilon_{ijk}P_k\,, \qquad [K_i, P_j] = iM\delta_{ij}\,, \\
[J_i, W] = [P_i, W] = 0\,, \qquad [K_i, W] = iP_i\,, \\
[J_i, M] = [P_i, M] = [K_i, M] = [W, M] = 0\,,
\end{gathered}$$

with K of order $1/v$. Note that the product of a translation $\mathbf{x} \to \mathbf{x} + \mathbf{a}$ and a 'boost' $\mathbf{x} \to \mathbf{x} + \mathbf{v}t$ should be the transformation $\mathbf{x} \to \mathbf{x} + \mathbf{v}t + \mathbf{a}$, but this is not true for the action of these operators on Hilbert space:

$$\exp(i\mathbf{K} \cdot \mathbf{v}) \exp(-i\mathbf{P} \cdot \mathbf{a}) = \exp(iM\mathbf{a} \cdot \mathbf{v}/2) \exp(i(\mathbf{K} \cdot \mathbf{v} - \mathbf{P} \cdot \mathbf{a}))\,.$$

The appearance of the phase factor $\exp(iM\mathbf{a} \cdot \mathbf{v}/2)$ shows that this is a projective representation, with a superselection rule forbidding the superposition of states of different mass. In this respect, the mathematics of the Poincaré group is simpler than that of the Galilean group. However, there is nothing to prevent us from formally enlarging the Galilean group, by adding one more generator to its Lie algebra, which commutes with all the other generators, and whose eigenvalues are the masses of the various states. In this case physical states provide an ordinary rather than a projective representation of the expanded symmetry group. The difference appears to be a mere matter of notation, except that with this reinterpretation of the Galilean group there is no need for a mass superselection rule.

2.5 One-Particle States

We now consider the classification of one-particle states according to their transformation under the inhomogeneous Lorentz group.

The components of the energy-momentum four-vector all commute with each other, so it is natural to express physical state-vectors in terms of eigenvectors of the four-momentum. Introducing a label σ to denote all other degrees of freedom, we thus consider state-vectors $|p,\sigma\rangle$ with

$$P^\mu |p,\sigma\rangle = p^\mu |p,\sigma\rangle \,. \tag{2.5.1}$$

For general states, consisting for instance of several unbound particles, the label σ would have to be allowed to include continuous as well as discrete labels. We take as part of the definition of a *one-particle* state, that the label σ is purely discrete, and will limit ourselves here to that case. (However, a specific bound state of two or more particles, such as the lowest state of the hydrogen atom, is to be considered as a one-particle state. It is not an *elementary* particle, but the distinction between composite and elementary particles is of no relevance here.)

Eqs. (2.5.1) and (2.4.26) tell us how the states $|p,\sigma\rangle$ transform under translations:

$$U(\mathbf{1},a)|p,\sigma\rangle = e^{-ip\cdot a}|p,\sigma\rangle \,.$$

We must now consider how these states transform under homogeneous Lorentz transformations.

Using (2.4.9), we see that the effect of operating on $|p,\sigma\rangle$ with a quantum homogeneous Lorentz transformation $U(\Lambda,0)\equiv U(\Lambda)$ is to produce an eigenvector of the four-momentum with eigenvalue Λp

$$\begin{aligned} P^\mu U(\Lambda)|p,\sigma\rangle &= U(\Lambda)\left[U^{-1}(\Lambda)P^\mu U(\Lambda)\right]|p,\sigma\rangle \\ &= U(\Lambda)(\Lambda^{-1\mu}{}_\rho P^\rho)|p,\sigma\rangle \\ &= \Lambda^\mu{}_\rho p^\rho U(\Lambda)|p,\sigma\rangle \,. \end{aligned} \tag{2.5.2}$$

Hence $U(\Lambda)|p,\sigma\rangle$ must be a linear combination of the state-vectors $|\Lambda p,\sigma'\rangle$:

$$U(\Lambda)|p,\sigma\rangle = \sum_{\sigma'} C_{\sigma'\sigma}(\Lambda,p)|\Lambda p,\sigma'\rangle \,. \tag{2.5.3}$$

In general, it may be possible by using suitable linear combinations of the $|p,\sigma\rangle$ to choose the σ labels in such a way that the matrix $C_{\sigma'\sigma}(\Lambda,p)$ is block-diagonal; in other words, so that the $|p,\sigma\rangle$ with σ within any one block by themselves furnish a representation of the inhomogeneous Lorentz group. It is natural to identify the states of a specific particle type with the components of a representation of the inhomogeneous Lorentz group which is irreducible, in the sense that it cannot be further decomposed in this way.[5] Our task now

[5]Of course, different particle species may correspond to representations that are isomorphic, i.e., that have matrices $C_{\sigma'\sigma}(\Lambda,p)$ that are either identical, or identical up to a similarity transformation. In some cases it may be convenient to define particle types as irreducible representations of larger groups that contain the inhomogeneous proper orthochronous Lorentz group as a subgroup; for instance, as we shall see, for massless particles whose interactions respect the symmetry of space inversion it is customary to treat all the components of an irreducible representation of the inhomogeneous Lorentz group including space inversion as a single particle type.

is to work out the structure of the coefficients $C_{\sigma'\sigma}(\Lambda, p)$ in irreducible representations of the inhomogeneous Lorentz group.

For this purpose, note that the only functions of p^μ that are left invariant by all proper orthochronous Lorentz transformations $\Lambda^\mu{}_\nu$, are the invariant square $p^2 \equiv \eta_{\mu\nu} p^\mu p^\nu$, and for $p^2 \leq 0$, also the sign of p^0. Hence, for each value of p^2, and (for $p^2 \leq 0$) each sign of p^0, we can choose a 'standard' four-momentum, say k^μ, and express any p^μ of this class as

$$p^\mu = L^\mu{}_\nu(p) k^\nu \,, \tag{2.5.4}$$

where $L^\mu{}_\nu$ is some standard Lorentz transformation that depends on p^μ, and also implicitly on our choice of the standard k^μ. We can then *define* the states $|p, \sigma\rangle$ of momentum p by

$$|p, \sigma\rangle \equiv N(p) U(L(p)) |k, \sigma\rangle \,, \tag{2.5.5}$$

where $N(p)$ is a numerical normalization factor, to be chosen later. Up to this point, we have said nothing about how the σ labels are related for different momenta; Eq. (2.5.5) now fills that gap.

Operating on (2.5.5) with an arbitrary homogeneous Lorentz transformation $U(\Lambda)$, we now find

$$\begin{aligned} U(\Lambda)|p, \sigma\rangle &= N(p) U(\Lambda L(p)) |k, \sigma\rangle \\ &= N(p) U(L(\Lambda p)) U(L^{-1}(\Lambda p) \Lambda L(p)) |k, \sigma\rangle \,. \end{aligned} \tag{2.5.6}$$

The point of this last step is that the Lorentz transformation $L^{-1}(\Lambda p)\Lambda L(p)$ takes k to $L(p)k = p$, and then to Λp, and then back to k, so it belongs to the subgroup of the homogeneous Lorentz group consisting of Lorentz transformations $W^\mu{}_\nu$ that leave k^μ invariant:

$$W^\mu{}_\nu k^\nu = k^\mu \,. \tag{2.5.7}$$

This subgroup is called the *little group* [5]. For any W satisfying Eq. (2.5.7), we have

$$U(W)|k, \sigma\rangle = \sum_{\sigma'} D_{\sigma'\sigma}(W) |k, \sigma'\rangle \,. \tag{2.5.8}$$

The coefficients $D(W)$ furnish a representation of the little group; that is, for any elements $\bar{W}, W$ we have

$$\begin{aligned} \sum_{\sigma'} D_{\sigma'\sigma}(\bar{W}W)|k, \sigma'\rangle &= U(\bar{W}W)|k, \sigma\rangle = U(\bar{W})U(W)|k, \sigma\rangle \\ &= U(\bar{W}) \sum_{\sigma''} D_{\sigma''\sigma}(W)|k, \sigma''\rangle \\ &= \sum_{\sigma'\sigma''} D_{\sigma'\sigma''}(\bar{W}) D_{\sigma''\sigma}(W) |k, \sigma'\rangle \end{aligned}$$

and so

$$D_{\sigma'\sigma}(\bar{W}W) = \sum_{\sigma''} D_{\sigma'\sigma''}(\bar{W}) D_{\sigma''\sigma}(W) \,. \tag{2.5.9}$$

In particular, we may apply Eq. (2.5.8) to the little-group transformation

$$W(\Lambda, p) \equiv L^{-1}(\Lambda p) \Lambda L(p) \tag{2.5.10}$$

and then Eq. (2.5.6) takes the form

$$U(\Lambda)|p,\sigma\rangle = N(p)\sum_{\sigma'} D_{\sigma'\sigma}(W(\Lambda,p))U(L(\Lambda p))|k,\sigma'\rangle\,,$$

or, recalling the definition (2.5.5):

$$U(\Lambda)|p,\sigma\rangle = \left(\frac{N(p)}{N(\Lambda p)}\right)\sum_{\sigma'} D_{\sigma'\sigma}(W(\Lambda,p))|\Lambda p,\sigma'\rangle\,. \tag{2.5.11}$$

Apart from the question of normalization, the problem of determining the coefficients $C_{\sigma'\sigma}$ in the transformation rule (2.5.3) has been reduced to the problem of finding the representations of the little group. This approach, of deriving representations of a group like the inhomogeneous Lorentz group from the representations of a little group, is called the method of induced representations [6].

Table 2.1 gives a convenient choice of the standard momentum k^μ and the corresponding little group for the various classes of four-momenta.

Of these six classes of four-momenta, only (a), (c), and (f) have any known interpretations in terms of physical states. Not much needs to be said here about case (f)—$p^\mu = 0$; it describes the vacuum, which is simply left invariant by $U(\Lambda)$. In what follows we will consider only cases (a) and (c), which cover particles of mass $M > 0$ and mass zero, respectively.

This is a good place to pause, and say something about the normalization of these states. By the usual orthonormalization procedure of quantum mechanics, we may choose the states with standard momentum k^μ to be orthonormal, in the sense that

$$\langle k',\sigma'|k,\sigma\rangle = \delta^3(\mathbf{k}'-\mathbf{k})\delta_{\sigma'\sigma}\,. \tag{2.5.12}$$

(The delta function appears here because $|k,\sigma\rangle$ and $|k',\sigma'\rangle$ are eigenstates of a Hermitian operator with eigenvalues $\mathbf{k}$ and $\mathbf{k}'$, respectively.) This has the immediate consequence that the representation of the little group in Eqs. (2.5.8) and (2.5.11) must be unitary[6]

$$D^\dagger(W) = D^{-1}(W)\,. \tag{2.5.13}$$

[6]The little groups $SO(2,1)$ and $SO(3,1)$ for $p^2 > 0$ and $p^\mu = 0$ have no non-trivial finite-dimensional unitary representations, so if there were any states with a given momentum p^μ with $p^2 > 0$ or $p^\mu = 0$ that transform non-trivially under the little group, there would have to be an infinite number of them.

Table 2.1. Standard momenta and the corresponding little group for various classes of four-momenta. Here κ is an arbitrary positive energy, say 1 eV. The little groups are mostly pretty obvious: $SO(3)$ is the ordinary rotation group in three dimensions (excluding space inversions), because rotations are the only proper orthochronous Lorentz transformations that leave at rest a particle with zero momentum, while $SO(2,1)$ and $SO(3,1)$ are the Lorentz groups in $(2+1)$- and $(3+1)$-dimensions, respectively. The group $ISO(2)$ is the group of Euclidean geometry, consisting of rotations and translations in two dimensions. Its appearance as the little group for $p^2 = 0$ is explained below.

	Standard k^μ	Little Group
(a) $p^2 = -M^2 < 0,\ p^0 > 0$	$(M, 0, 0, 0)$	$SO(3)$
(b) $p^2 = -M^2 < 0,\ p^0 < 0$	$(-M, 0, 0, 0)$	$SO(3)$
(c) $p^2 = 0,\ p^0 > 0$	$(\kappa, 0, 0, \kappa)$	$ISO(2)$
(d) $p^2 = 0,\ p^0 < 0$	$(-\kappa, 0, 0, \kappa)$	$ISO(2)$
(e) $p^2 = N^2 > 0$	$(0, 0, 0, N)$	$SO(2,1)$
(f) $p^\mu = 0$	$(0, 0, 0, 0)$	$SO(3,1)$

Now, what about scalar products for arbitrary momenta? Using the unitarity of the operator $U(\Lambda)$ in Eqs. (2.5.5) and (2.5.11), we find for the scalar product:

$$\begin{aligned}\langle p', \sigma' | p, \sigma \rangle &= N(p) \langle U^{-1}(L(p)) p', \sigma' | k, \sigma \rangle \\ &= N(p) N^*(p') D_{\sigma\sigma'}(W(L^{-1}(p), p'))^* \delta^3(\mathbf{k}' - \mathbf{k}) \,,\end{aligned}$$

where $k' \equiv L^{-1}(p)p'$. Since also $k = L^{-1}(p)p$, the delta function $\delta^3(\mathbf{k} - \mathbf{k}')$ is proportional to $\delta^3(\mathbf{p} - \mathbf{p}')$. For $p' = p$, the little-group transformation here is trivial, $W(L^{-1}(p), p) = \mathbf{1}$, and so the scalar product is

$$\langle p', \sigma' | p, \sigma \rangle = |N(p)|^2 \delta_{\sigma'\sigma} \delta^3(\mathbf{k}' - \mathbf{k}) \,. \tag{2.5.14}$$

It remains to work out the proportionality factor relating $\delta^3(\mathbf{k}-\mathbf{k}')$ and $\delta^3(\mathbf{p}-\mathbf{p}')$. Note that the Lorentz-invariant integral of an arbitrary scalar function $f(p)$ over four-momenta with $-p^2 = M^2 \geq 0$ and $p^0 > 0$ (i.e., cases (a) or (c)) may be written

$$\begin{aligned}\int d^4p\, \delta(p^2 + M^2) \theta(p^0) f(p) &= \int d^3\mathbf{p}\, dp^0\, \delta((p^0)^2 - \mathbf{p}^2 - M^2) \theta(p^0) f(\mathbf{p}, p^0) \\ &= \int d^3\mathbf{p}\, \frac{f(\mathbf{p}, \sqrt{\mathbf{p}^2 + M^2})}{2\sqrt{\mathbf{p}^2 + M^2}} \,.\end{aligned}$$

($\theta(p^0)$ is the step function: $\theta(x) = 1$ for $x \geq 0$, $\theta(x) = 0$ for $x < 0$.) We see that when integrating on the 'mass shell' $p^2 + M^2 = 0$, the invariant volume element is

$$d^3\mathbf{p} / \sqrt{\mathbf{p}^2 + M^2} \,. \tag{2.5.15}$$

The delta function is defined by

$$\begin{aligned}F(\mathbf{p}) &= \int F(\mathbf{p}')\delta^3(\mathbf{p}-\mathbf{p}')d^3\mathbf{p}' \\ &= \int F(\mathbf{p}')\left[\sqrt{\mathbf{p}'^2+M^2}\,\delta^3(\mathbf{p}'-\mathbf{p})\right]\frac{d^3\mathbf{p}'}{\sqrt{\mathbf{p}'^2+M^2}}\,,\end{aligned}$$

so we see that the invariant delta function is

$$\sqrt{\mathbf{p}'^2+M^2}\,\delta^3(\mathbf{p}'-\mathbf{p}) = p^0\delta^3(\mathbf{p}'-\mathbf{p})\,. \tag{2.5.16}$$

Since p' and p are related to k' and k respectively by a Lorentz transformation, $L(p)$, we have then

$$p^0\delta^3(\mathbf{p}'-\mathbf{p}) = k^0\delta^3(\mathbf{k}'-\mathbf{k})$$

and therefore

$$\langle p',\sigma'|p,\sigma\rangle = |N(p)|^2\delta_{\sigma'\sigma}\left(\frac{p^0}{k^0}\right)\delta^3(\mathbf{p}'-\mathbf{p})\,. \tag{2.5.17}$$

The normalization factor $N(p)$ is sometimes chosen to be just $N(p)=1$, but then we would need to keep track of the p^0/k^0 factor in scalar products. Instead, I will here adopt the more usual convention that

$$N(p) = \sqrt{k^0/p^0} \tag{2.5.18}$$

for which

$$\langle p',\sigma'|p,\sigma\rangle = \delta_{\sigma'\sigma}\delta^3(\mathbf{p}'-\mathbf{p})\,. \tag{2.5.19}$$

We now consider the two cases of physical interest: particles of mass $M>0$, and particles of zero mass.

Mass Positive-Definite

The little group here is the three-dimensional rotation group. Its unitary representations can be broken up into a direct sum of irreducible unitary representations [7] $D^{(j)}_{\sigma'\sigma}(R)$ of dimensionality $2j+1$, with $j=0,\frac{1}{2},1,\ldots$. These can be built up from the standard matrices for infinitesimal rotations $R_{ik}=\delta_{ik}+\Theta_{ik}$, with $\Theta_{ik}=-\Theta_{ki}$ infinitesimal:

$$D^{(j)}_{\sigma'\sigma}(\mathbf{1}+\Theta) = \delta_{\sigma'\sigma} + \frac{i}{2}\Theta_{ik}(J^{(j)}_{ik})_{\sigma'\sigma}\,, \tag{2.5.20}$$

$$\begin{aligned}(J^{(j)}_{23}\pm iJ^{(j)}_{31})_{\sigma'\sigma} &= (J^{(j)}_{1}\pm iJ^{(j)}_{2})_{\sigma'\sigma} \\ &= \delta_{\sigma',\sigma\pm1}\sqrt{(j\mp\sigma)(j\pm\sigma+1)}\,,\end{aligned} \tag{2.5.21}$$

$$(J^{(j)}_{12})_{\sigma'\sigma} = (J^{(j)}_{3})_{\sigma'\sigma} = \sigma\delta_{\sigma'\sigma}\,, \tag{2.5.22}$$

with σ running over the values $j,j-1,\ldots,-j$. For a particle of mass $M>0$ and spin j, Eq. (2.5.11) now becomes

$$U(\Lambda)|p,\sigma\rangle = \sqrt{\frac{(\Lambda p)^0}{p^0}}\sum_{\sigma'}D^{(j)}_{\sigma'\sigma}(W(\Lambda,p))|\Lambda p,\sigma'\rangle\,, \tag{2.5.23}$$

with the little-group element $W(\Lambda, p)$ (the Wigner rotation [5]) given by Eq. (2.5.10):

$$W(\Lambda, p) = L^{-1}(\Lambda p)\Lambda L(p) \,.$$

To calculate this rotation, we need to choose a 'standard boost' $L(p)$ which carries the four-momentum from $k^\mu = (M, \mathbf{0})$ to p^μ. This is conveniently chosen as

$$\begin{aligned} L^i{}_k(p) &= \delta_{ik} + (\gamma - 1)\hat{p}_i \hat{p}_k \,, \\ L^i{}_0(p) &= L^0{}_i(p) = \hat{p}_i \sqrt{\gamma^2 - 1} \,, \\ L^0{}_0(p) &= \gamma \,, \end{aligned} \tag{2.5.24}$$

where

$$\hat{p}_i \equiv p_i / |\mathbf{p}| \,, \qquad \gamma \equiv \sqrt{\mathbf{p}^2 + M^2}/M \,.$$

It is very important that when $\Lambda^\mu{}_\nu$ is an arbitrary three-dimensional rotation $\mathcal{R}$, the Wigner rotation $W(\Lambda, p)$ is the same as $\mathcal{R}$ for all p. To see this, note that the boost (2.5.24) may be expressed as

$$L(p) = R(\hat{\mathbf{p}}) B(|\mathbf{p}|) R^{-1}(\hat{\mathbf{p}}) \,,$$

where $R(\hat{\mathbf{p}})$ is a rotation (to be defined in a standard way below, in Eq. (2.5.47)) that takes the three-axis into the direction of $\mathbf{p}$, and

$$B(|\mathbf{p}|) = \begin{pmatrix} \gamma & 0 & 0 & \sqrt{\gamma^2 - 1} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \sqrt{\gamma^2 - 1} & 0 & 0 & \gamma \end{pmatrix} .$$

Then for an arbitrary rotation $\mathcal{R}$

$$W(\mathcal{R}, p) = R(\mathcal{R}\hat{\mathbf{p}}) B^{-1}(|\mathbf{p}|) R^{-1}(\mathcal{R}\hat{\mathbf{p}}) \mathcal{R} R(\hat{\mathbf{p}}) B(|\mathbf{p}|) R^{-1}(\hat{\mathbf{p}}) \,.$$

But the rotation $R^{-1}(\mathcal{R}\hat{\mathbf{p}})\mathcal{R}R(\hat{\mathbf{p}})$ takes the three-axis into the direction $\hat{\mathbf{p}}$, and then into the direction $\mathcal{R}\hat{\mathbf{p}}$, and then back to the three-axis, so it must be just a rotation by some angle θ around the three-axis

$$R^{-1}(\mathcal{R}\hat{\mathbf{p}})\mathcal{R}R(\hat{\mathbf{p}}) = R(\theta) \equiv \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & \sin\theta & 0 \\ 0 & -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} .$$

Since $R(\theta)$ commutes with $B(|\mathbf{p}|)$, this now gives

$$\begin{aligned} W(\mathcal{R}, p) &= R(\mathcal{R}\hat{\mathbf{p}}) B^{-1}(|\mathbf{p}|) R(\theta) B(|\mathbf{p}|) R^{-1}(\hat{\mathbf{p}}) \\ &= R(\mathcal{R}\hat{\mathbf{p}}) R(\theta) R^{-1}(\hat{\mathbf{p}}) \end{aligned}$$

and hence

$$W(\mathcal{R}, p) = \mathcal{R}$$

as was to be shown. Thus states of a moving massive particle (and, by extension, multi-particle states) have the same transformation under rotations as in non-relativistic quantum mechanics. This is another piece of good news—the whole apparatus of spherical harmonics, Clebsch–Gordan coefficients, etc. can be carried over wholesale from non-relativistic to relativistic quantum mechanics.

Mass Zero

First, we have to work out the structure of the little group. Consider an arbitrary little-group element $W^\mu{}_\nu$, with $W^\mu{}_\nu k^\nu = k^\mu$, where k^μ is the standard four-momentum for this case, $k^\mu = (1,0,0,1)$. Acting on a time-like four-vector $t^\mu = (1,0,0,0)$, such a Lorentz transformation must yield a four-vector Wt whose length and scalar product with $Wk = k$ are the same as those of t:

$$(Wt)^\mu (Wt)_\mu = t^\mu t_\mu = -1\,, \qquad (Wt)^\mu k_\mu = t^\mu k_\mu = -1\,.$$

Any four-vector that satisfies the second condition may be written

$$(Wt)^\mu = (1+\zeta, \alpha, \beta, \zeta)$$

and the first condition then yields the relation

$$\zeta = (\alpha^2 + \beta^2)/2\,. \tag{2.5.25}$$

It follows that the effect of $W^\mu{}_\nu$ on t^ν is the same as that of the Lorentz transformation

$$S^\mu{}_\nu(\alpha,\beta) = \begin{pmatrix} 1+\zeta & \alpha & \beta & -\zeta \\ \alpha & 1 & 0 & -\alpha \\ \beta & 0 & 1 & -\beta \\ \zeta & \alpha & \beta & 1-\zeta \end{pmatrix}. \tag{2.5.26}$$

This does not mean that W equals $S(\alpha,\beta)$, but it does mean that $S^{-1}(\alpha,\beta)W$ is a Lorentz transformation that leaves the time-like four-vector $(1,0,0,0)$ invariant, and is therefore a pure rotation. Also, $S^\mu{}_\nu$ like $W^\mu{}_\nu$ leaves the light-like four-vector $(1,0,0,1)$ invariant, so $S^{-1}(\alpha,\beta)W$ must be a rotation by some angle θ around the three-axis

$$S^{-1}(\alpha,\beta)W = R(\theta)\,, \tag{2.5.27}$$

where

$$R^\mu{}_\nu(\theta) \equiv \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & \sin\theta & 0 \\ 0 & -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

The most general element of the little group is therefore of the form

$$W(\theta,\alpha,\beta) = S(\alpha,\beta)R(\theta)\,. \tag{2.5.28}$$

What group is this? We note that the transformations with $\theta = 0$ or with $\alpha = \beta = 0$ form subgroups:

$$S(\bar\alpha,\bar\beta)S(\alpha,\beta) = S(\bar\alpha+\alpha, \bar\beta+\beta) \tag{2.5.29}$$

$$R(\bar\theta)R(\theta) = R(\bar\theta+\theta)\,. \tag{2.5.30}$$

These subgroups are *Abelian*—that is, their elements all commute with each other. Furthermore, the subgroup with $\theta = 0$ is *invariant*, in the sense that its elements are transformed into other elements of the same subgroup by any member of the group

$$R(\theta)S(\alpha,\beta)R^{-1}(\theta) = S(\alpha\cos\theta + \beta\sin\theta, -\alpha\sin\theta + \beta\cos\theta)\,. \tag{2.5.31}$$

From Eqs. (2.5.29)–(2.5.31) we can work out the product of any group elements. The reader will recognize these multiplication rules as those of the group $ISO(2)$, consisting of translations (by a vector (α,β)) and rotations (by an angle θ) in two dimensions.

Groups that do *not* have invariant Abelian subgroups have certain simple properties, and for this reason are called *semi-simple*. As we have seen, the little group $ISO(2)$ like the inhomogeneous Lorentz group is *not* semi-simple, and this leads to interesting complications. First, let's take a look at the Lie algebra of $ISO(2)$. For θ,α,β infinitesimal, the general group element is

$$W(\theta,\alpha,\beta)^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu\,, \qquad \omega_{\mu\nu} = \begin{pmatrix} 0 & -\alpha & -\beta & 0 \\ \alpha & 0 & \theta & -\alpha \\ \beta & -\theta & 0 & -\beta \\ 0 & \alpha & \beta & 0 \end{pmatrix}.$$

From (2.4.3), we see then that the corresponding Hilbert space operator is

$$U(W(\theta,\alpha,\beta)) = \mathbf{1} + i\alpha A + i\beta B + i\theta J_3\,, \tag{2.5.32}$$

where A and B are the Hermitian operators

$$A = -J^{13} + J^{10} = J_2 + K_1\,, \tag{2.5.33}$$

$$B = -J^{23} + J^{20} = -J_1 + K_2\,, \tag{2.5.34}$$

and, as before, $J_3 = J_{12}$. Either from (2.4.18)–(2.4.20), or directly from Eqs. (2.5.29)–(2.5.31), we see that these generators have the commutators

$$[J_3, A] = +iB\,, \tag{2.5.35}$$

$$[J_3, B] = -iA\,, \tag{2.5.36}$$

$$[A, B] = 0\,. \tag{2.5.37}$$

Since A and B are commuting Hermitian operators they (like the momentum generators of the inhomogeneous Lorentz group) can be simultaneously diagonalized by states $|k,a,b\rangle$

$$A|k,a,b\rangle = a|k,a,b\rangle\,, \qquad B|k,a,b\rangle = b|k,a,b\rangle\,.$$

The problem is that if we find one such set of non-zero eigenvalues of A, B, then we find a whole continuum. From Eq. (2.5.31), we have

$$U[R(\theta)]A\,U^{-1}[R(\theta)] = A\cos\theta - B\sin\theta\,,$$

$$U[R(\theta)]B\,U^{-1}[R(\theta)] = A\sin\theta + B\cos\theta\,,$$

and so, for arbitrary θ,

$$A|k,a,b\rangle^\theta = (a\cos\theta - b\sin\theta)|k,a,b\rangle^\theta \,,$$

$$B|k,a,b\rangle^\theta = (a\sin\theta + b\cos\theta)|k,a,b\rangle^\theta \,,$$

where

$$|k,a,b\rangle^\theta \equiv U^{-1}(R(\theta))|k,a,b\rangle \,.$$

Massless particles are not observed to have any continuous degree of freedom like θ; to avoid such a continuum of states, we must require that physical states (now called $|k,\sigma\rangle$) are eigenvectors of A and B with $a = b = 0$:

$$A|k,\sigma\rangle = B|k,\sigma\rangle = 0 \,. \tag{2.5.38}$$

These states are then distinguished by the eigenvalue of the remaining generator

$$J_3|k,\sigma\rangle = \sigma|k,\sigma\rangle \,. \tag{2.5.39}$$

Since the momentum $\mathbf{k}$ is in the three-direction, σ gives the component of angular momentum in the direction of motion, or *helicity*.

We are now in a position to calculate the Lorentz transformation properties of general massless particle states. First note that by use of the general arguments of Section 2.2, Eq. (2.5.32) generalizes for finite α and β to

$$U(S(\alpha,\beta)) = \exp(i\alpha A + i\beta B) \tag{2.5.40}$$

and for finite θ to

$$U(R(\theta)) = \exp(iJ_3\theta) \,. \tag{2.5.41}$$

An arbitrary element W of the little group can be put in the form (2.5.28), so that

$$U(W)|k,\sigma\rangle = \exp(i\alpha A + i\beta B)\exp(i\theta J_3)|k,\sigma\rangle = \exp(i\theta\sigma)|k,\sigma\rangle$$

and therefore Eq. (2.5.8) gives

$$D_{\sigma'\sigma}(W) = \exp(i\theta\sigma)\delta_{\sigma'\sigma} \,,$$

where θ is the angle defined by expressing W as in Eq. (2.5.28). The Lorentz transformation rule for a massless particle of arbitrary helicity is now given by Eqs. (2.5.11) and (2.5.18) as

$$U(\Lambda)|p,\sigma\rangle = \sqrt{\frac{(\Lambda p)^0}{p^0}}\exp(i\sigma\theta(\Lambda,p))|\Lambda p,\sigma\rangle \tag{2.5.42}$$

with $\theta(\Lambda,p)$ defined by

$$W(\Lambda,p) \equiv L^{-1}(\Lambda p)\Lambda L(p) \equiv S(\alpha(\Lambda,p),\beta(\Lambda,p))R(\theta(\Lambda,p)) \,. \tag{2.5.43}$$

We shall see in Section 5.9 that electromagnetic gauge invariance arises from the part of the little group parameterized by α and β.

At this point we have not yet encountered any reason that would forbid the helicity σ of a massless particle from being an arbitrary real number. As we shall see in Section 2.7, there are topological considerations that restrict the allowed values of σ to integers and half-integers, just as for massive particles.

To calculate the little-group element (2.5.43) for a given Λ and p, (and also to enable us to calculate the effect of space or time inversion on these states in the next section) we need to fix a convention for the standard Lorentz transformation that takes us from $k^\mu = (\kappa, 0, 0, \kappa)$ to p^μ. This may conveniently be chosen to have the form

$$L(p) = R(\hat{\mathbf{p}})B(|\mathbf{p}|/\kappa) \tag{2.5.44}$$

where $B(u)$ is a pure boost along the three-direction:

$$B(u) \equiv \begin{pmatrix} (u^2+1)/2u & 0 & 0 & (u^2-1)/2u \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ (u^2-1)/2u & 0 & 0 & (u^2+1)/2u \end{pmatrix} \tag{2.5.45}$$

and $R(\hat{\mathbf{p}})$ is a pure rotation that carries the three-axis into the direction of the unit vector $\hat{\mathbf{p}}$. For instance, suppose we take $\hat{\mathbf{p}}$ to have polar and azimuthal angles θ and ϕ:

$$\hat{\mathbf{p}} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)\,. \tag{2.5.46}$$

Then we can take $R(\hat{\mathbf{p}})$ as a rotation by angle θ around the two-axis, which takes $(0,0,1)$ into $(\sin\theta, 0, \cos\theta)$, followed by a rotation by angle ϕ around the three-axis:

$$U(R(\hat{\mathbf{p}})) = \exp(i\phi J_3)\exp(i\theta J_2)\,, \tag{2.5.47}$$

where $0 \le \theta \le \pi$, $0 \le \phi < 2\pi$. (We give $U(R(\hat{\mathbf{p}}))$ rather than $R(\hat{\mathbf{p}})$, together with a specification of the range of ϕ and θ, because shifting θ or ϕ by 2π would give the same rotation $R(\hat{\mathbf{p}})$, but a different sign for $U(R(\hat{\mathbf{p}}))$ when acting on half-integer spin states.) Since (2.5.47) is a rotation, and does take the three-axis into the direction (2.5.46), any other choice of such an $R(\hat{\mathbf{p}})$ would differ from this one by at most an initial rotation around the three-axis, corresponding to a mere redefinition of the phase of the one-particle states.

Note that the helicity is Lorentz-invariant; a massless particle of a given helicity σ looks the same (aside from its momentum) in all inertial frames. Indeed, we would be justified in thinking of massless particles of each different helicity as different species of particles. However, as we shall see in the next section, particles of opposite helicity are related by the symmetry of space inversion. Thus, because electromagnetic and gravitational forces obey space inversion symmetry, the massless particles of helicity ± 1 associated with electromagnetic phenomena are both called *photons*, and the massless particles of helicity ± 2 that are believed to be associated with gravitation are both called *gravitons*. On the other hand, the supposedly massless particles of helicity $\pm 1/2$ that are emitted in nuclear beta decay have no interactions (apart from gravitation) that respect the symmetry of space inversion, so these particles are given different names: *neutrinos* for helicity $+1/2$, and *antineutrinos* for helicity $-1/2$.

Even though the helicity of a massless particle is Lorentz-invariant, the state itself is not. In particular, because of the helicity-dependent phase factor $\exp(i\sigma\theta)$ in Eq. (2.5.42), a state formed as a linear superposition of one-particle states with opposite helicities will be changed by a Lorentz transformation into a different superposition. For instance, a general one-photon state of four-momenta may be written

$$|p;\alpha\rangle = \alpha_+|p,+1\rangle + \alpha_-|p,-1\rangle\,,$$

where

$$|\alpha_+|^2 + |\alpha_-|^2 = 1\,.$$

The generic case is one of *elliptic polarization*, with $|\alpha_\pm|$ both non-zero and unequal. *Circular polarization* is the limiting case where either α_+ or α_- vanishes, and *linear polarization* is the opposite extreme, with $|\alpha_+| = |\alpha_-|$. The overall phase of α_+ and α_- has no physical significance, and for linear polarization may be adjusted so that $\alpha_- = \alpha_+^*$, but the relative phase is still important. Indeed, for linear polarizations with $\alpha_- = \alpha_+^*$, the phase of α_+ may be identified as the angle between the plane of polarization and some fixed reference direction perpendicular to $\mathbf{p}$. Eq. (2.5.42) shows that under a Lorentz transformation $\Lambda^\mu{}_\nu$, this angle rotates by an amount $\theta(\Lambda, p)$. Plane polarized gravitons can be defined in a similar way, and here Eq. (2.5.42) has the consequence that a Lorentz transformation Λ rotates the plane of polarization by an angle $2\theta(\Lambda, p)$.

2.6 Space Inversion and Time-Reversal

We saw in Section 2.3 that any homogeneous Lorentz transformation is either proper and orthochronous (i.e., $\mathrm{Det}\,\Lambda = +1$ and $\Lambda^0{}_0 \geq +1$) or else equal to a proper orthochronous transformation times either $\mathcal{P}$ or $\mathcal{T}$ or $\mathcal{PT}$, where $\mathcal{P}$ and $\mathcal{T}$ are the space inversion and time-reversal transformations

$$\mathcal{P}^\mu{}_\nu = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \qquad \mathcal{T}^\mu{}_\nu = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

It used to be thought self-evident that the fundamental multiplication rule of the Poincaré group

$$U(\bar\Lambda, \bar a)U(\Lambda, a) = U(\bar\Lambda\Lambda, \bar\Lambda a + \bar a)$$

would be valid even if Λ and/or $\bar\Lambda$ involved factors of $\mathcal{P}$ or $\mathcal{T}$ or $\mathcal{PT}$. In particular, it was believed that there are operators corresponding to $\mathcal{P}$ and $\mathcal{T}$ themselves:

$$P \equiv U(\mathcal{P}, 0), \qquad T \equiv U(\mathcal{T}, 0)$$

such that

$$PU(\Lambda, a)P^{-1} = U(\mathcal{P}\Lambda\mathcal{P}^{-1}, \mathcal{P}a), \tag{2.6.1}$$

$$TU(\Lambda, a)T^{-1} = U(\mathcal{T}\Lambda\mathcal{T}^{-1}, \mathcal{T}a) \tag{2.6.2}$$

for any proper orthochronous Lorentz transformation $\Lambda^{\mu}{}_{\nu}$, and translation a^{μ}. These transformation rules incorporate most of what is meant when we say that P or T are 'conserved'.

In 1956–57 it became understood [8] that this is true for P only in the approximation in which one ignores the effects of weak interactions, such as those that produce nuclear beta decay. Time-reversal survived for a while, but in 1964 there appeared indirect evidence [9] that these properties of T are also only approximately satisfied. (See Section 3.3.) In what follows, we will make believe that operators P and T satisfying Eqs. (2.6.1) and (2.6.2) actually exist, but it should be kept in mind that this is only an approximation.

Let us apply Eqs. (2.6.1) and (2.6.2) in the case of an infinitesimal transformation, i.e.,

$$\Lambda^{\mu}{}_{\nu} = \delta^{\mu}{}_{\nu} + \omega^{\mu}{}_{\nu}, \qquad a^{\mu} = \epsilon^{\mu}$$

with $\omega_{\mu\nu} = -\omega_{\nu\mu}$ and ϵ_{μ} both infinitesimal. Using (2.4.3), and equating coefficients of $\omega_{\rho\sigma}$ and ϵ_{ρ} in Eqs. (2.6.1) and (2.6.2), we obtain the P and T transformation properties of the Poincaré generators

$$PiJ^{\rho\sigma}P^{-1} = i\mathcal{P}_{\mu}{}^{\rho}\mathcal{P}_{\nu}{}^{\sigma}J^{\mu\nu}, \tag{2.6.3}$$

$$PiP^{\rho}P^{-1} = i\mathcal{P}_{\mu}{}^{\rho}P^{\mu}, \tag{2.6.4}$$

$$TiJ^{\rho\sigma}T^{-1} = i\mathcal{T}_{\mu}{}^{\rho}\mathcal{T}_{\nu}{}^{\sigma}J^{\mu\nu}, \tag{2.6.5}$$

$$TiP^{\rho}T^{-1} = i\mathcal{T}_{\mu}{}^{\rho}P^{\mu}. \tag{2.6.6}$$

This is much like Eqs. (2.4.8) and (2.4.9), except that we have not cancelled factors of i on both sides of these equations, because at this point we have not yet decided whether P and T are linear and unitary or antilinear and antiunitary.

The decision is an easy one. Setting $\rho = 0$ in Eq. (2.6.4) gives

$$PiHP^{-1} = iH,$$

where $H \equiv P^0$ is the energy operator. If P were antiunitary and antilinear then it would anticommute with i, so $PHP^{-1} = -H$. But then for any state $|\Psi\rangle$ of energy $E > 0$, there would have to be another state $P^{-1}|\Psi\rangle$ of energy $-E < 0$. There are no states of negative energy (energy less than that of the vacuum), so we are forced to choose the other alternative: *P is linear and unitary, and commutes rather than anticommutes with H.*

On the other hand, setting $\rho = 0$ in Eq. (2.6.6) yields

$$TiHT^{-1} = -iH.$$

If we supposed that T is linear and unitary then we could simply cancel the is, and find $THT^{-1} = -H$, with the again disastrous conclusion that for any state $|\Psi\rangle$ of energy E there is another state $T^{-1}|\Psi\rangle$ of energy $-E$. To avoid this, we are forced here to conclude that *T is antilinear and antiunitary.*

Now that we have decided that P is linear and T is antilinear, we can conveniently rewrite Eqs. (2.6.3)–(2.6.6) in terms of the generators (2.4.15)–(2.4.17) in a three-dimensional notation

$$PJP^{-1} = +\mathbf{J}, \tag{2.6.7}$$
$$P\mathbf{K}P^{-1} = -\mathbf{K}, \tag{2.6.8}$$
$$P\mathbf{P}P^{-1} = -\mathbf{P}, \tag{2.6.9}$$
$$T\mathbf{J}T^{-1} = -\mathbf{J}, \tag{2.6.10}$$
$$T\mathbf{K}T^{-1} = +\mathbf{K}, \tag{2.6.11}$$
$$T\mathbf{P}T^{-1} = -\mathbf{P}, \tag{2.6.12}$$

and, as shown before,

$$PHP^{-1} = THT^{-1} = H. \tag{2.6.13}$$

It is physically sensible that P should preserve the sign of $\mathbf{J}$, because at least the orbital part is a vector product $\mathbf{r} \times \mathbf{p}$ of two vectors, both of which change sign under an inversion of the spatial coordinate system. On the other hand, T reverses $\mathbf{J}$, because after time-reversal an observer will see all bodies spinning in the opposite direction. Note by the way that Eq. (2.6.10) is consistent with the angular-momentum commutation relations $\mathbf{J} \times \mathbf{J} = i\mathbf{J}$, because T reverses not only $\mathbf{J}$, but also i. The reader can easily check that Eqs. (2.6.7)–(2.6.13) are consistent with all the commutation relations (2.4.18)–(2.4.24).

Let us now consider what P and T do to one-particle states:

$P : M > 0$

The one-particle states $|k, \sigma\rangle$ are defined as eigenvectors of $\mathbf{P}$, H, and J_3 with eigenvalues 0, M, and σ, respectively. From Eqs. (2.6.7), (2.6.9), and (2.6.13), we see that the same must be true of the state $P|k, \sigma\rangle$, and therefore (barring degeneracies) these states can only differ by a phase

$$P|k, \sigma\rangle = \eta_\sigma |k, \sigma\rangle$$

with a phase factor ($|\eta_\sigma| = 1$) that may or may not depend on the spin σ. To see that η_σ is σ-independent, we note from (2.5.8), (2.5.20), and (2.5.21) that

$$(J_1 \pm iJ_2)|k, \sigma\rangle = \sqrt{(j \mp \sigma)(j \pm \sigma + 1)}|k, \sigma \pm 1\rangle, \tag{2.6.14}$$

where j is the particle's spin. Operating on both sides with P, we find

$$\eta_\sigma = \eta_{\sigma \pm 1}$$

and so η_σ is actually independent of σ. We therefore write

$$P|k, \sigma\rangle = \eta |k, \sigma\rangle \tag{2.6.15}$$

with η a phase, known as the *intrinsic parity*, that depends only on the species of particle on which P acts.

To get to finite momentum states, we must apply the unitary operator $U(L(p))$ corresponding to the 'boost' (2.5.24):

$$|p, \sigma\rangle = \sqrt{M/p^0}\, U(L(p))|k, \sigma\rangle.$$

We note that

$$\mathcal{P}L(p)\mathcal{P}^{-1} = L(\mathcal{P}p), \qquad \mathcal{P}p = (p^0, -\mathbf{p}),$$

so using Eqs. (2.6.1) and (2.6.15), we have

$$P|p,\sigma\rangle = \sqrt{M/p^0}\, U(L(\mathcal{P}p))\eta|k,\sigma\rangle$$

or in other words

$$P|p,\sigma\rangle = \eta|\mathcal{P}p,\sigma\rangle. \tag{2.6.16}$$

$T : M > 0$

From Eqs. (2.6.10), (2.6.12), and (2.6.13), we see that the effect of T on the zero-momentum one-particle state $|k,\sigma\rangle$ is to yield a state with

$$\begin{aligned}
\mathbf{P}(T|k,\sigma\rangle) &= 0, \\
H(T|k,\sigma\rangle) &= M(T|k,\sigma\rangle), \\
J_3(T|k,\sigma\rangle) &= -\sigma(T|k,\sigma\rangle),
\end{aligned}$$

and so

$$T|k,\sigma\rangle = \zeta_\sigma|k,-\sigma\rangle,$$

where ζ_σ is a phase factor. Applying the operator T to (2.6.14), and recalling that T anticommutes with $\mathbf{J}$ *and* i, we find

$$(-J_1 \pm iJ_2)\zeta_\sigma|k,-\sigma\rangle = \sqrt{(j\mp\sigma)(j\pm\sigma+1)}\,\zeta_{\sigma\pm1}|k,-\sigma\mp1\rangle.$$

Using Eq. (2.6.14) again on the left, we see that the square-root factors cancel, and so

$$-\zeta_\sigma = \zeta_{\sigma\pm1}.$$

We write the solution as $\zeta_\sigma = \zeta(-1)^{j-\sigma}$, with ζ another phase that depends only on the species of particle:

$$T|k,\sigma\rangle = \zeta(-1)^{j-\sigma}|k,-\sigma\rangle. \tag{2.6.17}$$

However, unlike the 'intrinsic parity' η, the time-reversal phase ζ has no physical significance. This is because we can redefine the one-particle states by a change of phase

$$|k,\sigma\rangle \longrightarrow |k,\sigma\rangle' = \zeta^{1/2}|k,\sigma\rangle$$

in such a way that the phase ζ is eliminated from the transformation rule

$$T|k,\sigma\rangle' = \zeta^{*1/2}T|k,\sigma\rangle = \zeta^{*1/2}\zeta(-1)^{j-\sigma}|k,-\sigma\rangle = (-1)^{j-\sigma}|k,-\sigma\rangle'.$$

In what follows we will keep the arbitrary phase ζ in Eq. (2.6.17), just to keep open our options in choosing the phase of the one-particle states, but it should be kept in mind that this phase is of no real importance.

To deal with states of finite momentum, we again apply the 'boost' (2.5.24). Note that

$$\mathcal{T}L(p)\mathcal{T}^{-1} = L(\mathcal{P}p), \qquad \mathcal{P}p = (p^0, -\mathbf{p}).$$

(That is, changing the sign of each element of $L^\mu{}_\nu$ with an odd number of *time*-indices is the same as changing the signs of elements with an odd number of *space*-indices.) Using Eqs. (2.6.2) and (2.5.5), we have then

$$T|p,\sigma\rangle = \zeta(-1)^{j-\sigma}|\mathcal{P}p,-\sigma\rangle. \tag{2.6.18}$$

$P : M = 0$

Acting on a state $|k,\sigma\rangle$ that is defined as an eigenvector of P^μ with eigenvalue $k^\mu = (\kappa,0,0,\kappa)$ and an eigenvector of J_3 with eigenvalue σ, the parity operator P yields a state with four-momentum $(\mathcal{P}k)^\mu = (\kappa,0,0,-\kappa)$ and J_3 equal to σ. Thus it takes a state of helicity (the component of spin along the direction of motion) σ into one of helicity $-\sigma$. As mentioned earlier, this shows that the existence of a space-inversion symmetry requires that any species of massless particle with non-zero helicity must be accompanied with another of opposite helicity. Because P does not leave the standard momentum invariant, it is convenient to consider instead the operator $U(R_2^{-1})P$, where R_2 is a rotation that also takes k to $\mathcal{P}k$, conveniently chosen as a rotation by $-180°$ around the two-axis

$$U(R_2) = \exp(-i\pi J_2). \tag{2.6.19}$$

Since $U(R_2^{-1})$ reverses the sign of J_3, we have

$$U(R_2^{-1})P|k,\sigma\rangle = \eta_\sigma|k,-\sigma\rangle \tag{2.6.20}$$

with η_σ a phase factor. Now, $R_2^{-1}\mathcal{P}$ commutes with the Lorentz 'boost' (2.5.45), and $\mathcal{P}$ commutes with the rotation $R(\hat{\mathbf{p}})$ which takes the three-direction into the direction of $\mathbf{p}$, so by operating on (2.5.5) with P, we find for a general four-momentum p^μ

$$\begin{aligned} P|p,\sigma\rangle &= \sqrt{\frac{\kappa}{p^0}}\,U\left(R(\hat{\mathbf{p}})R_2 B\left(\frac{|\mathbf{p}|}{\kappa}\right)\right)U(R_2^{-1})P|k,\sigma\rangle \\ &= \sqrt{\frac{\kappa}{p^0}}\,\eta_\sigma U\left(R(\hat{\mathbf{p}})R_2 B\left(\frac{|\mathbf{p}|}{\kappa}\right)\right)|k,-\sigma\rangle. \end{aligned}$$

Note that $R(\hat{\mathbf{p}})R_2$ is a rotation that takes the three-axis into the direction of $-\hat{\mathbf{p}}$, but $U(R(\hat{\mathbf{p}})R_2)$ is not quite equal to $U(R(-\hat{\mathbf{p}}))$. According to (2.5.47),

$$U(R(-\hat{\mathbf{p}})) = \exp\big(i(\phi \pm \pi)J_3\big)\exp\big(i(\pi-\theta)J_2\big)$$

with azimuthal angle chosen as $\phi+\pi$ or $\phi-\pi$ according to whether $0 \le \phi < \pi$ or $\pi \le \phi < 2\pi$, so that it remains in the range of 0 to 2π. Then

$$\begin{aligned} U^{-1}(R(-\hat{\mathbf{p}}))U(R(\hat{\mathbf{p}})R_2) &= \exp\big(-i(\pi-\theta)J_2\big) \\ &\quad\times \exp\big(-i(\phi\pm\pi)J_3\big)\exp(i\phi J_3)\exp(i\theta J_2)\exp(-i\pi J_2) \\ &= \exp\big(-i(\pi-\theta)J_2\big)\exp(\mp i\pi J_3)\exp\big(-i(\pi-\theta)J_2\big). \end{aligned}$$

But a rotation of $\pm 180°$ around the three-axis reverses the sign of J_2, so

$$U(R(\hat{\mathbf{p}})R_2) = U(R(-\hat{\mathbf{p}}))\exp(\pm i\pi J_3). \tag{2.6.21}$$

Also, $R(-\hat{\mathbf{p}})B(|\mathbf{p}|/\kappa)$ is just the standard boost $L(\mathcal{P}p)$ in the direction $\mathcal{P}p = (p^0, -\mathbf{p})$. We have then finally

$$P|p,\sigma\rangle = \eta_\sigma \exp(\mp i\pi\sigma)|\mathcal{P}p, -\sigma\rangle \tag{2.6.22}$$

with the phase $-\pi\sigma$ or $+\pi\sigma$ according to whether the two-component of $\mathbf{p}$ is positive or negative, respectively. This peculiar change of sign in the operation of parity for massless particles of half-integer spin is due to the convention adopted in Eq. (2.5.47) for the rotation used to define massless particle states of arbitrary momentum. Because the rotation group is not simply connected, some discontinuity of this sort is unavoidable.

$T : M = 0$

Acting on the state $|k,\sigma\rangle$, which has values $k^\mu = (\kappa, 0, 0, \kappa)$ and σ for P^μ and J_3, the time-reversal operator T yields a state which has values $(\mathcal{P}k)^\mu = (\kappa, 0, 0, -\kappa)$ and $-\sigma$ for P^μ and J_3. Thus T does not change the helicity $\mathbf{J}\cdot\hat{\mathbf{k}}$, and by itself has nothing to say about whether massless particles of one helicity σ are accompanied with others of helicity $-\sigma$. Because T like P does not leave the standard four-momentum k invariant, it is convenient to consider the generator $U(R_2^{-1})T$, where R_2 is the rotation (2.6.19), which also takes k into $\mathcal{P}k$. This commutes with J_3, so

$$U(R_2^{-1})T|k,\sigma\rangle = \zeta_\sigma|k,\sigma\rangle \tag{2.6.23}$$

with ζ_σ another phase. Since $R_2^{-1}\mathcal{T}$ commutes with the boost (2.5.45), and $\mathcal{T}$ commutes with the rotation $R(\hat{\mathbf{p}})$, operating with T on the state (2.5.5) gives

$$T|p,\sigma\rangle = \sqrt{\frac{\kappa}{p^0}}\, U\left[R(\hat{\mathbf{p}})R_2 B\left(\frac{|\mathbf{p}|}{\kappa}\right)\right]\zeta_\sigma|k,\sigma\rangle. \tag{2.6.24}$$

Using Eq. (2.6.21), this yields finally

$$T|p,\sigma\rangle = \zeta_\sigma \exp(\pm i\pi\sigma)|\mathcal{P}p,\sigma\rangle. \tag{2.6.25}$$

Again, the top or bottom sign applies according to whether the two-component of $\mathbf{p}$ is positive or negative, respectively.

* * *

It is interesting that the square T^2 of the time-reversal operator has a very simple action on both massive and massless one-particle states. Using Eq. (2.6.18), and recalling that T is antiunitary, we see that for massive one-particle states:

$$\begin{aligned} T^2|p,\sigma\rangle &= T\zeta(-1)^{j-\sigma}|\mathcal{P}p,-\sigma\rangle \\ &= \zeta^*(-1)^{j-\sigma}\zeta(-1)^{j+\sigma}|p,\sigma\rangle \end{aligned}$$

or in other words

$$T^2|p,\sigma\rangle = (-1)^{2j}|p,\sigma\rangle. \tag{2.6.26}$$

We get the same result for massless particles. If the two-component of $\mathbf{p}$ is positive then the two-component of $\mathcal{P}\mathbf{p}$ is negative, and vice-versa, so Eq. (2.6.25) gives

$$\begin{aligned}T^2|p,\sigma\rangle &= T\zeta_\sigma \exp(\pm i\pi\sigma)|\mathcal{P}p,\sigma\rangle \\ &= \zeta_\sigma^* \exp(\mp i\pi\sigma)\zeta_\sigma \exp(\mp i\pi\sigma)|p,\sigma\rangle \\ &= \exp(\mp 2\pi i\sigma)|p,\sigma\rangle.\end{aligned}$$

As long as σ is an integer or half-integer, this can be written

$$T^2|p,\sigma\rangle = (-1)^{2|\sigma|}|p,\sigma\rangle. \tag{2.6.27}$$

By the 'spin' of a massless particle, we usually mean the absolute value of the helicity, so Eq. (2.6.27) is the same as Eq. (2.6.26).

This result has an interesting consequence. When T^2 acts on any state $|\Psi\rangle$ of a system of non-interacting particles, either massive or massless, it yields a factor $(-1)^{2j}$ or $(-1)^{2|\sigma|}$ for each particle. Hence if the state contains an odd number of particles of half-integer spin or helicity (plus any number of particles of integer spin or helicity), we get an overall change of sign

$$T^2|\Psi\rangle = -|\Psi\rangle. \tag{2.6.28}$$

If we now 'turn on' various interactions, this result will be preserved, provided these interactions respect invariance under time-reversal, even if they do not respect rotational invariance. (For instance, these arguments will apply even if our system is subjected to arbitrary static gravitational and electric fields.) Now, suppose that $|\Psi\rangle$ is an eigenstate of the Hamiltonian. Since T commutes with the Hamiltonian, $T|\Psi\rangle$ will also be an eigenstate of the Hamiltonian. Is it the same state? If so, then $T|\Psi\rangle$ can differ from $|\Psi\rangle$ only by a phase

$$T|\Psi\rangle = \zeta|\Psi\rangle,$$

but then

$$T^2|\Psi\rangle = T(\zeta|\Psi\rangle) = \zeta^* T|\Psi\rangle = |\zeta|^2|\Psi\rangle = |\Psi\rangle,$$

in contradiction with Eq. (2.6.28). We see that any energy eigenstate $|\Psi\rangle$ satisfying Eq. (2.6.28) must be degenerate, with another eigenstate of the same energy. This is known as a 'Kramers degeneracy' [10]. Of course, this conclusion is trivial if the system is in a rotationally invariant environment, because the total angular-momentum j of any state of this system would have to be a half-integer, and there would therefore be $2j + 1 = 2, 4, \ldots$ degenerate states. The surprising result is that at least a two-fold degeneracy persists even if rotational invariance is perturbed by external fields, such as electrostatic fields, as long as these fields are invariant under T. In particular, if any particle had an electric or gravitational dipole moment then the degeneracy among its $2j + 1$ spin states would be entirely removed in a static electric or gravitational field, so such dipole moments are forbidden by time-reversal invariance.

For the sake of completeness, it should be mentioned that P and T can have more complicated effects on multiplets of particles with the same mass. This possibility will be considered in Appendix C of this chapter. No physically relevant examples are known.

2.7 Projective Representations[7]

We now return to the possibility mentioned in Section 2.2, that a group of symmetries may be represented projectively on physical states; that is, the elements T, $\bar{T}$, etc. of the symmetry group may be represented on the physical Hilbert space by unitary operators $U(T)$, $U(\bar{T})$, etc., which satisfy the composition rule

$$U(T)U(\bar{T}) = \exp\bigl(i\phi(T,\bar{T})\bigr)U(T\bar{T}) \tag{2.7.1}$$

with ϕ a real phase. (A bar is used here just to distinguish one symmetry operator from another.) The basic requirement that any phase ϕ in Eq. (2.7.1) would have to satisfy is the associativity condition

$$U(T_3)\bigl(U(T_2)U(T_1)\bigr) = \bigl(U(T_3)U(T_2)\bigr)U(T_1),$$

which imposes on ϕ the corresponding condition

$$\phi(T_2,T_1) + \phi(T_3,T_2T_1) = \phi(T_3,T_2) + \phi(T_3T_2,T_1). \tag{2.7.2}$$

Of course, any phase of the form

$$\phi(T,\bar{T}) = \alpha(T\bar{T}) - \alpha(T) - \alpha(\bar{T}) \tag{2.7.3}$$

will automatically satisfy Eq. (2.7.2), but a projective representation with such a phase can be replaced with an ordinary representation by replacing $U(T)$ with

$$\widetilde{U}(T) \equiv U(T)\exp\bigl(i\alpha(T)\bigr)$$

for which

$$\widetilde{U}(T)\widetilde{U}(\bar{T}) = \widetilde{U}(T\bar{T}).$$

Any set of functions $\phi(T,\bar{T})$ that satisfy Eq. (2.7.2), and that differ only by functions $\Delta\phi(T,\bar{T})$ of the form (2.7.3), is called a 'two-cocycle'. A trivial cocycle is one that contains the function $\phi = 0$, and hence consists of functions of the form (2.7.3), which can be eliminated by a redefinition of $U(T)$. We are interested here in whether a symmetry group allows any non-trivial two-cocyles; that is, whether it may have a representation on the physical Hilbert space that is *intrinsically* projective, in the sense that the phase $\phi(T,\bar{T})$ *cannot* be eliminated in this way.

In order to answer this question, it is useful first to consider the effect of a phase ϕ in Eq. (2.7.1) on the commutation relations of the generators of infinitesimal transformations. When either $\bar{T}$ or T is the identity, the phase ϕ must clearly vanish

$$\phi(T,\mathbf{1}) = \phi(\mathbf{1},\bar{T}) = 0. \tag{2.7.4}$$

[7] This section lies somewhat out of the book's main line of development, and may be omitted in a first reading.

When both T and $\bar{T}$ are *near* the identity the phase must be small. Using coordinates θ^a to parameterize group elements (as in Section 2.2), with $T(0) = \mathbf{1}$, Eq. (2.7.4) tells us that the expansion of $\phi[T(\theta), T(\bar{\theta})]$ around $\theta = \bar{\theta} = 0$ must start with terms of order $\theta\bar{\theta}$:

$$\phi\big(T(\theta), T(\bar{\theta})\big) = f_{ab}\theta^a\bar{\theta}^b + \cdots, \tag{2.7.5}$$

where f_{ab} are real numerical constants. Inserting this expansion in the power series expansion of Eq. (2.7.1), and repeating the steps that led to (2.2.22), we now have

$$[t_b, t_c] = iC^a{}_{bc}t_a + iC_{bc}\mathbf{1}, \tag{2.7.6}$$

where C_{bc} is the antisymmetric coefficient

$$C_{bc} = -f_{bc} + f_{cb}. \tag{2.7.7}$$

The appearance of terms on the right-hand side of the commutation relation proportional to the unit element (so-called *central charges*) is the counterpart for the Lie algebra of the presence of phases in a projective representation of a group.

The constants C_{bc} as well as $C^a{}_{bc}$ are subject to an important constraint, which follows from the Jacobi identity. Taking the commutator of (2.7.6) with t_d, and adding the same expressions with b, c, d replaced with c, d, b and d, b, c, the sum of the three double-commutators on the left-hand side vanishes identically, and so

$$C^a{}_{bc}C^e{}_{ad} + C^a{}_{cd}C^e{}_{ab} + C^a{}_{db}C^e{}_{ac} = 0 \tag{2.7.8}$$

and also

$$C^a{}_{bc}C_{ad} + C^a{}_{cd}C_{ab} + C^a{}_{db}C_{ac} = 0. \tag{2.7.9}$$

Eq. (2.7.9) always has one obvious class of non-zero solutions for C_{ab}:

$$C_{ab} = C^e{}_{ab}\phi_e, \tag{2.7.10}$$

where ϕ_e is an arbitrary set of real constants. For these solutions, we can eliminate the central charges from Eq. (2.7.6) by a redefinition of the generators

$$t_a \longrightarrow \widetilde{t}_a \equiv t_a + \phi_a. \tag{2.7.11}$$

The new generators then satisfy the commutation relations without central charges

$$[\widetilde{t}_b, \widetilde{t}_c] = iC^a{}_{bc}\widetilde{t}_a. \tag{2.7.12}$$

A given Lie algebra may or may not allow solutions of Eq. (2.7.9) other than Eq. (2.7.10).

We can now state the key theorem that governs the occurrence of intrinsically projective representations. The phase of any representation $U(T)$ of a given group can be chosen so that $\phi = 0$ in Eq. (2.7.1), if two conditions are met:

(a) The generators of the group in this representation can be redefined (as in Eq. (2.7.11)), so as to eliminate all central charges from the Lie algebra.

(b) The group is simply connected, i.e., any two group elements may be connected by a path lying within the group, and any two such paths may be continuously transformed into one another. (An equivalent statement is that any loop that starts and ends at the same group element may be shrunk continuously to a point.)

This theorem is proved in Appendix B of this chapter, which also offers comments about the case of groups that are not simply connected. It shows that there are just two (not exclusive) ways that intrinsically projective representations may arise: either algebraically, because the group is represented projectively even near the identity, or topologically, because the group is not simply connected, and hence a path from $\mathbf{1}$ to T and then from T to $\bar{T}$ may not be continuously deformable into some other path from $\mathbf{1}$ to $T\bar{T}$. In the latter case, the phase ϕ in Eq. (2.7.1) depends on the particular choice of standard paths, leading from the origin to the various group elements, that are used to define the corresponding U-operators.

Let's now consider each of these possibilities in turn for the special case of the inhomogeneous Lorentz group.

(A) Algebra

With central charges, the commutation relations of the generators of the inhomogeneous Lorentz group would read

$$i[J^{\mu\nu}, J^{\rho\sigma}] = \eta^{\nu\rho}J^{\mu\sigma} - \eta^{\mu\rho}J^{\nu\sigma} - \eta^{\sigma\mu}J^{\rho\nu} + \eta^{\sigma\nu}J^{\rho\mu} + C^{\rho\sigma,\mu\nu}, \tag{2.7.13}$$
$$i[P^{\mu}, J^{\rho\sigma}] = \eta^{\mu\rho}P^{\sigma} - \eta^{\mu\sigma}P^{\rho} + C^{\rho\sigma,\mu}, \tag{2.7.14}$$
$$i[J^{\mu\nu}, P^{\rho}] = \eta^{\nu\rho}P^{\mu} - \eta^{\mu\rho}P^{\nu} + C^{\rho,\mu\nu}, \tag{2.7.15}$$
$$i[P^{\mu}, P^{\rho}] = C^{\rho,\mu} \tag{2.7.16}$$

in place of Eqs. (2.4.12)–(2.4.14). We see that the Cs also satisfy the antisymmetry conditions

$$C^{\rho\sigma,\mu\nu} = -C^{\mu\nu,\rho\sigma}, \tag{2.7.17}$$
$$C^{\rho\sigma,\mu} = -C^{\mu,\rho\sigma}, \tag{2.7.18}$$
$$C^{\rho,\mu} = -C^{\mu,\rho}. \tag{2.7.19}$$

We will now show that all these constants have additional algebraic properties that allow them to be eliminated by shifting the definitions of $J^{\mu\nu}$ and P^{μ} by constant terms. (This corresponds to redefining the phase of the operators $U(\Lambda, a)$.) To derive these properties, we apply the Jacobi identities

$$[J^{\mu\nu}, [P^{\rho}, P^{\sigma}]] + [P^{\sigma}, [J^{\mu\nu}, P^{\rho}]] + [P^{\rho}, [P^{\sigma}, J^{\mu\nu}]] = 0, \tag{2.7.20}$$
$$[J^{\lambda\eta}, [J^{\mu\nu}, P^{\rho}]] + [P^{\rho}, [J^{\lambda\eta}, J^{\mu\nu}]] + [J^{\mu\nu}, [P^{\rho}, J^{\lambda\eta}]] = 0, \tag{2.7.21}$$
$$[J^{\lambda\eta}, [J^{\mu\nu}, J^{\rho\sigma}]] + [J^{\rho\sigma}, [J^{\lambda\eta}, J^{\mu\nu}]] + [J^{\mu\nu}, [J^{\rho\sigma}, J^{\lambda\eta}]] = 0. \tag{2.7.22}$$

(The Jacobi identity involving three Ps is automatically satisfied, and hence yields no further information.) Using Eqs. (2.7.13)–(2.7.16) in Eqs. (2.7.20)–(2.7.22), we obtain algebraic conditions on the Cs

$$0 = \eta^{\nu\rho} C^{\mu,\sigma} - \eta^{\mu\rho} C^{\nu,\sigma} - \eta^{\nu\sigma} C^{\mu,\rho} + \eta^{\mu\sigma} C^{\nu,\rho}, \tag{2.7.23}$$

$$\begin{aligned} 0 = {} & \eta^{\nu\rho} C^{\mu,\lambda\eta} - \eta^{\mu\rho} C^{\nu,\lambda\eta} - \eta^{\mu\lambda} C^{\rho,\eta\nu} + \eta^{\mu\eta} C^{\rho,\lambda\nu} \\ & + \eta^{\lambda\nu} C^{\rho,\mu\eta} - \eta^{\eta\nu} C^{\rho,\mu\lambda} + \eta^{\rho\lambda} C^{\eta,\mu\nu} - \eta^{\rho\eta} C^{\lambda,\mu\nu}, \end{aligned} \tag{2.7.24}$$

$$\begin{aligned} 0 = {} & \eta^{\nu\rho} C^{\mu\sigma,\lambda\eta} - \eta^{\mu\rho} C^{\nu\sigma,\lambda\eta} - \eta^{\sigma\mu} C^{\rho\nu,\lambda\eta} + \eta^{\sigma\nu} C^{\rho\mu,\lambda\eta} \\ & + \eta^{\eta\mu} C^{\lambda\nu,\rho\sigma} - \eta^{\lambda\mu} C^{\eta\nu,\rho\sigma} - \eta^{\nu\lambda} C^{\mu\eta,\rho\sigma} + \eta^{\nu\eta} C^{\mu\lambda,\rho\sigma} \\ & + \eta^{\sigma\lambda} C^{\rho\eta,\mu\nu} - \eta^{\rho\lambda} C^{\sigma\eta,\mu\nu} - \eta^{\sigma\eta} C^{\rho\lambda,\mu\nu} + \eta^{\rho\eta} C^{\sigma\lambda,\mu\nu}. \end{aligned} \tag{2.7.25}$$

Contracting Eq. (2.7.23) with $\eta_{\nu\rho}$ gives

$$C^{\mu,\sigma} = 0. \tag{2.7.26}$$

On the other hand, the constants $C^{\mu,\lambda\eta}$ and $C^{\rho\sigma,\mu\nu}$ are not necessarily zero, but their algebraic structure is simple enough so that they can be eliminated by shifting the definitions of P^μ and $J^{\mu\nu}$, respectively. Contracting Eq. (2.7.24) with $\eta_{\nu\rho}$ gives

$$C^{\mu,\lambda\eta} = \eta^{\eta\mu} C^{\lambda} - \eta^{\mu\lambda} C^{\eta}, \tag{2.7.27}$$

$$C^{\lambda} \equiv \frac{1}{3} \eta_{\rho\nu} C^{\rho,\lambda\nu}. \tag{2.7.28}$$

Also, contracting (2.7.25) with $\eta_{\nu\rho}$ gives

$$C^{\mu\sigma,\lambda\eta} = \eta^{\eta\mu} C^{\lambda\sigma} - \eta^{\lambda\mu} C^{\eta\sigma} + \eta^{\sigma\lambda} C^{\eta\mu} - \eta^{\eta\sigma} C^{\lambda\mu}, \tag{2.7.29}$$

$$C^{\lambda\sigma} \equiv \frac{1}{2} \eta_{\nu\rho} C^{\lambda\nu,\sigma\rho}. \tag{2.7.30}$$

(These expressions automatically satisfy Eqs. (2.7.24) and (2.7.25), so there is no further information to be gained from the Jacobi identities.) We now see that if the Cs are not zero, they can be eliminated by defining new generators

$$\widetilde{P}^\mu \equiv P^\mu + C^\mu, \tag{2.7.31}$$

$$\widetilde{J}^{\mu\sigma} \equiv J^{\mu\sigma} + C^{\mu\sigma}, \tag{2.7.32}$$

and the commutation relations are then what they would be for an ordinary representation

$$i[\widetilde{J}^{\mu\nu}, \widetilde{J}^{\rho\sigma}] = \eta^{\nu\rho} \widetilde{J}^{\mu\sigma} - \eta^{\mu\rho} \widetilde{J}^{\nu\sigma} - \eta^{\sigma\mu} \widetilde{J}^{\rho\nu} + \eta^{\sigma\nu} \widetilde{J}^{\rho\mu}, \tag{2.7.33}$$

$$i[\widetilde{J}^{\mu\nu}, \widetilde{P}^{\rho}] = \eta^{\nu\rho} \widetilde{P}^{\mu} - \eta^{\mu\rho} \widetilde{P}^{\nu}, \tag{2.7.34}$$

$$i[\widetilde{P}^{\mu}, \widetilde{P}^{\rho}] = 0. \tag{2.7.35}$$

The commutation relations will always be taken in the form Eqs. (2.7.33)–(2.7.35), but with tildas dropped.

Incidentally, the fact that there is no central charge in the algebra of the $J^{\mu\nu}$ could have been immediately inferred from the fact that this algebra is of the type known as 'semi-simple'. (Semi-simple Lie algebras are those that have no 'invariant Abelian' subalgebra, consisting of generators that commute with each other and whose commutators with any other generators also belong to the subalgebra.) There is a general theorem [11] that any central charges in semi-simple Lie algebras may always be removed by a redefinition of generators, as in Eq. (2.7.32). On the other hand, the full Poincaré algebra spanned by $J^{\mu\nu}$ and P^μ is not semi-simple (the P^μ form an invariant Abelian subalgebra), and we needed a special argument to show that its central charges can also be eliminated in this way. Indeed, the non-semi-simple Galilean algebra discussed in Section 2.4 does allow a central charge, the mass M.

We see that the inhomogeneous Lorentz group satisfies the first of the two conditions needed to rule out intrinsically projective representations. How about the second?

(B) Topology

To explore the topology of the inhomogeneous Lorentz group, it is very convenient to represent homogeneous Lorentz transformations by 2×2 complex matrices. Any real four-vector V^μ can be used to construct an Hermitian 2×2 matrix

$$v \equiv V^\mu \sigma_\mu = \begin{pmatrix} V^0 + V^3 & V^1 - iV^2 \\ V^1 + iV^2 & V^0 - V^3 \end{pmatrix}, \tag{2.7.36}$$

where σ_μ are the usual Pauli matrices with $\sigma_0 = \mathbf{1}$. Conversely any 2×2 Hermitian matrix can be put in this form, and therefore defines a real four-vector V^μ.

The property of Hermiticity will be preserved under the transformation

$$v \longrightarrow \lambda v \lambda^\dagger \tag{2.7.37}$$

with λ an arbitrary complex 2×2 matrix. Furthermore, the covariant square of the four-vector is

$$V_\mu V^\mu = (V^1)^2 + (V^2)^2 + (V^3)^2 - (V^0)^2 = -\operatorname{Det} v \tag{2.7.38}$$

and this determinant is preserved by the transformation (2.7.37) provided that

$$|\operatorname{Det} \lambda| = 1. \tag{2.7.39}$$

Each complex 2×2 matrix λ satisfying Eq. (2.7.39) thus defines a real linear transformation of V^μ that leaves Eq. (2.7.38) invariant, i.e., a homogeneous Lorentz transformation $\Lambda(\lambda)$:

$$\lambda V^\mu \sigma_\mu \lambda^\dagger = \big(\Lambda^\mu{}_\nu(\lambda) V^\nu\big)\sigma_\mu. \tag{2.7.40}$$

Furthermore, for two such matrices λ and $\bar\lambda$, we have

$$\begin{aligned}(\lambda\bar\lambda)V^\mu\sigma_\mu(\lambda\bar\lambda)^\dagger &= \lambda\big(\bar\lambda V^\mu \sigma_\mu \bar\lambda^\dagger\big)\lambda^\dagger \\ &= \lambda\Lambda^\mu{}_\nu(\bar\lambda)V^\nu\sigma_\mu\lambda^\dagger \\ &= \Lambda^\mu{}_\rho(\lambda)\Lambda^\rho{}_\nu(\bar\lambda)V^\nu\sigma_\mu\end{aligned}$$

and so

$$\Lambda(\lambda\bar{\lambda}) = \Lambda(\lambda)\Lambda(\bar{\lambda}). \tag{2.7.41}$$

However, two λs that differ only by an overall phase have the same effect on v in Eq. (2.7.37), and so correspond to the same Lorentz transformation. It is therefore convenient to adjust the phase of the λs so that

$$\mathrm{Det}\,\lambda = 1, \tag{2.7.42}$$

which is consistent with Eq. (2.7.41). The 2×2 complex matrices with unit determinant form a group, known as $SL(2, \mathbb{C})$. (The SL stands for 'special linear', with 'special' denoting a unit determinant, while C stands for 'complex'.) The group elements depend on $4-1 = 3$ complex parameters, or 6 real parameters, the same number as the Lorentz group. However, $SL(2, \mathbb{C})$ is not the same as the Lorentz group; if λ is a matrix in $SL(2, \mathbb{C})$, then so is $-\lambda$, and both λ and $-\lambda$ produce the same Lorentz transformation in Eq. (2.7.37). Indeed, it is easy to see that the matrix

$$\lambda(\theta) = \begin{pmatrix} e^{i\theta/2} & 0 \\ 0 & e^{-i\theta/2} \end{pmatrix}$$

produces a Lorentz transformation $\Lambda(\lambda(\theta))$ which is just a rotation by an angle θ around the three-axis, and hence $\lambda = -\mathbf{1}$ produces a rotation by an angle 2π. The Lorentz group is not the same as $SL(2, \mathbb{C})$, but rather[8] $SL(2, \mathbb{C})/\mathbb{Z}_2$, which is the group of complex 2×2 matrices with unit determinant, and with λ identified with $-\lambda$.

Now, what is the topology of the Lorentz group? By the polar decomposition theorem [12], any complex non-singular matrix λ may be written in the form

$$\lambda = ue^h,$$

where u is unitary and h is Hermitian

$$u^\dagger u = \mathbf{1}, \qquad h^\dagger = h.$$

Since $\mathrm{Det}\,u$ is a phase factor, and $\mathrm{Det}\exp h = \exp \mathrm{Tr}\, h$ is real and positive, the condition (2.7.42) requires both

$$\mathrm{Det}\,u = 1, \qquad \mathrm{Tr}\,h = 0.$$

(The factor u simply provides the rotation subgroup of the Lorentz group; if u is unitary then $\mathrm{Tr}(uvu^\dagger) = \mathrm{Tr}\,v$, so $V^0 = \frac{1}{2}\mathrm{Tr}\,v$ is left invariant by $\Lambda(u)$.) Furthermore, this decomposition is unique, so $SL(2, \mathbb{C})$ is topologically just the direct product (i.e., the set of pairs of points) of the space of all us and the space of all hs. Any Hermitian traceless 2×2 matrix h can be expressed as

$$h = \begin{pmatrix} c & a - ib \\ a + ib & -c \end{pmatrix}$$

[8] The group $\mathbb{Z}_2$ consists of just elements $+\mathbf{1}$ and $-\mathbf{1}$. In general, when we write G/H, with H an invariant subgroup of G, we mean the group G with elements g and gh identified if $g \in G$ and $h \in H$. The subgroup $\mathbb{Z}_2$ is trivially invariant because its elements commute with all elements of $SL(2, \mathbb{C})$.

with a, b, c real but otherwise unconstrained, so the space of all hs is topologically the same as ordinary three-dimensional flat space, $\mathbb{R}^3$. On the other hand, any unitary 2×2 matrix with unit determinant can be expressed as

$$u = \begin{pmatrix} d + ie & f + ig \\ -f + ig & d - ie \end{pmatrix}$$

with d, e, f, g subject to the single non-linear constraint

$$d^2 + e^2 + f^2 + g^2 = 1,$$

so the space $SU(2)$ of all us is topologically the same as S^3, the three-dimensional surface of a spherical ball in flat four-dimensional space. Thus $SL(2, \mathbb{C})$ is topologically the same as the direct product $\mathbb{R}^3 \times S^3$. This is simply connected: any curve connecting two points of $\mathbb{R}^3$ or S^3 can be deformed into any other, and the same is true of the direct product. (All spheres S_n except the circle S_1 are simply connected.) However, we are interested in $SL(2, \mathbb{C})/\mathbb{Z}_2$, not $SL(2, \mathbb{C})$. Identifying λ with $-\lambda$ is the same as identifying the unitary factors u and $-u$ (because e^h is always positive), so the Lorentz group has the topology of $\mathbb{R}^3 \times S^3/\mathbb{Z}_2$, where $S^3/\mathbb{Z}_2$ is the three-dimensional spherical surface with opposite points of the sphere identified. This is *not* simply connected; for instance, a path on S^3 from u to u' cannot be continuously deformed into a path on S^3 from u to $-u'$, even though these two paths link the same points of $S^3/\mathbb{Z}_2$. In fact, $S^3/\mathbb{Z}_2$ is *doubly* connected; the paths between any two points fall into two classes, depending on whether or not they involve an inversion $u \to -u$, and any path in one class can be deformed into another path of that class. An equivalent statement is that a double loop, that goes twice over the same path from any element back to itself, may be continuously contracted to a point. (As discussed in Appendix B, this is summarized mathematically in the statement that the fundamental group, or first homotopy group, of $S^3/\mathbb{Z}_2$ is $\mathbb{Z}_2$.) Similarly, the inhomogeneous Lorentz group has the same topology as $\mathbb{R}^4 \times \mathbb{R}^3 \times S^3/\mathbb{Z}_2$, and is therefore also doubly connected.

Because the Lorentz group (homogeneous or inhomogeneous) is not simply connected, it does have intrinsically projective representations. However, because the double loop that goes twice from $\mathbf{1}$ to Λ to $\Lambda\bar{\Lambda}$ and then back to $\mathbf{1}$ can be contracted to a point, we must have

$$\left[U(\Lambda)U(\bar{\Lambda})U^{-1}(\Lambda\bar{\Lambda})\right]^2 = \mathbf{1}$$

and hence the phase $e^{i\phi(\Lambda,\bar{\Lambda})}$ is just a sign

$$U(\Lambda)U(\bar{\Lambda}) = \pm U(\Lambda\bar{\Lambda}). \tag{2.7.43}$$

Likewise, for the inhomogeneous Lorentz group

$$U(\Lambda, a)U(\bar{\Lambda}, \bar{a}) = \pm U(\Lambda\bar{\Lambda}, \Lambda\bar{a} + a). \tag{2.7.44}$$

These 'representations up to a sign' are familiar; they are just the states of integer spin, for which the signs in Eqs. (2.7.43) and (2.7.44) are always $+1$, and the states of half-integer spin, for which these signs are $+1$ or -1 according to whether the path from $\mathbf{1}$ to

Λ to $\Lambda\bar{\Lambda}$ and then back to $\mathbf{1}$ is or is not contractible to a point. This difference arises because a rotation of 2π around the three-axis acting on a state with angular-momentum three-component σ produces a phase $e^{2\pi i\sigma}$, and thus has no effect on a state of integer spin and produces a sign change when acting on a state of half-integer spin. (These two cases correspond to the two irreducible representations of the first homotopy group, $\mathbb{Z}_2$.) Thus Eq. (2.7.43) or Eq. (2.7.44) imposes a superselection rule: we must not mix states of integer and half-integer spin.

For finite mass, the limitation to integer or half-integer spin was previously derived by purely algebraic means from the well-known representations of the generators of the little group, which here are just the angular-momentum matrices $J^{(j)}$ with j integer or half-integer. On the other hand, for zero mass the action of the little group on physical one-particle states is just a rotation around the momentum, and here there is no *algebraic* reason for a limitation to integer or half-integer helicity. There is, however, a *topological* reason: a rotation by an angle 4π around the momentum can be continuously deformed into no rotation at all, so the factor $\exp(4\pi i\sigma)$ must be unity, and hence σ must be an integer or half-integer.

Instead of working with projective representations and imposing a superselection rule, we can just as well expand the Lorentz group, taking it as $SL(2,\mathbb{C})$ itself, instead of $SL(2,\mathbb{C})/\mathbb{Z}_2$ as before. Ordinary rotation invariance forbids transitions between states of integer and half-integer total spin, so the only difference is that now the group is simply-connected, and it therefore has only ordinary representations, not projective representations, so that we cannot infer a superselection rule. This does not mean that we actually can prepare physical systems in linear combinations of states of integer and half-integer spin, but only that the observed Lorentz invariance of nature cannot be used to show that such superpositions are impossible.

Similar remarks apply to any symmetry group. If its Lie algebra involves central charges, then we can always expand the algebra to include generators that commute with anything, and whose eigenvalues are the central charges, just as we did when we added a mass operator to the Lie algebra of the Galilean group at the end of Section 2.4. The expanded Lie algebra is then, of course, free of central charges, so the part of the group near the identity has only ordinary representations, and does not require any superselection rule. Likewise, even though a Lie group G may not be simply connected, it can always be expressed as C/H, where C is a simply connected group known as the 'universal covering group' of G, and H is an invariant subgroup[9] of C. In general, we may just as well take the symmetry group as C instead of G, because there is no difference in their consequences, except that G implies a superselection rule, while C does not. In short, the issue of superselection rules is a bit of a red herring; *it may or it may not be possible to prepare physical systems in arbitrary superpositions of states, but one cannot settle the question by reference to symmetry principles, because whatever one thinks the symmetry group of nature may be,*

[9]The first homotopy group of C/H is H. We have seen that the covering group of the homogeneous Lorentz group is $SL(2,\mathbb{C})$, and the covering group of the three-dimensional rotation group is $SU(2)$. This connection with SL and SU groups is special to the case of three, four, or six dimensions; for general dimensions d the covering group of $SO(d)$ is given a special name, 'Spin(d)'.

there is always another group whose consequences are identical except for the absence of superselection rules.

Appendix A The Symmetry Representation Theorem

This appendix presents the proof of the fundamental theorem of Wigner [2] that any symmetry transformation can be represented on the Hilbert space of physical states by an operator that is either linear and unitary or antilinear and antiunitary. For our present purposes, the property of symmetry transformations on which we chiefly rely is that they are ray transformations T that preserve transition probabilities, in the sense that if $|\Psi_1\rangle$ and $|\Psi_2\rangle$ are state-vectors belonging to rays $\mathcal{R}_1$ and $\mathcal{R}_2$ then any state-vectors $|\Psi'_1\rangle$ and $|\Psi'_2\rangle$ belonging to the transformed rays $T\mathcal{R}_1$ and $T\mathcal{R}_2$ satisfy

$$\left|\langle\Psi'_1|\Psi'_2\rangle\right|^2 = \left|\langle\Psi_1|\Psi_2\rangle\right|^2. \tag{2.A.1}$$

We also require that a symmetry transformation should have an inverse that preserves transition probabilities in the same sense.

To start, consider some complete orthonormal set of state-vectors $|\Psi_k\rangle$ belonging to rays $\mathcal{R}_k$, with

$$\langle\Psi_k|\Psi_l\rangle = \delta_{kl}, \tag{2.A.2}$$

and let $|\Psi'_k\rangle$ be some arbitrary choice of state-vectors belonging to the transformed rays $T\mathcal{R}_k$. From Eq. (2.A.1), we have

$$\left|\langle\Psi'_k|\Psi'_l\rangle\right|^2 = \left|\langle\Psi_k|\Psi_l\rangle\right|^2 = \delta_{kl}.$$

But $\langle\Psi'_k|\Psi'_k\rangle$ is automatically real and positive, so this requires that it should have the value unity, and therefore

$$\langle\Psi'_k|\Psi'_l\rangle = \delta_{kl}. \tag{2.A.3}$$

It is easy to see that these transformed states $|\Psi'_k\rangle$ also form a complete set, for if there were any non-zero state-vector $|\Psi'\rangle$ that was orthogonal to all of the $|\Psi'_k\rangle$, then the inverse transform of the ray to which $|\Psi'\rangle$ belongs would consist of non-zero state-vectors $|\Psi''\rangle$ for which, for all k:

$$\left|\langle\Psi_k|\Psi''\rangle\right|^2 = \left|\langle\Psi'_k|\Psi'\rangle\right|^2 = 0,$$

which is impossible since the $|\Psi_k\rangle$ were assumed to form a complete set.

We must now establish a phase convention for the states $|\Psi'_k\rangle$. For this purpose, we single out one of the $|\Psi_k\rangle$, say $|\Psi_1\rangle$, and consider the state-vectors

$$|Y_k\rangle = \frac{1}{\sqrt{2}}\big[|\Psi_1\rangle + |\Psi_k\rangle\big] \tag{2.A.4}$$

belonging to some ray $\mathcal{S}_k$, with $k \neq 1$. Any state-vector $|Y'_k\rangle$ belonging to the transformed ray $T\mathcal{S}_k$ may be expanded in the state-vectors $|\Psi'_l\rangle$,

$$|Y'_k\rangle = \sum_l c_{kl}|\Psi'_l\rangle.$$

From Eq. (2.A.1) we have

$$|c_{kk}| = |c_{k1}| = \frac{1}{\sqrt{2}}$$

and for $l \neq k$ and $l \neq 1$:

$$c_{kl} = 0.$$

For any given k, by an appropriate choice of phase of the two state-vectors $|Y_k'\rangle$ and $|\Psi_k'\rangle$, we can clearly adjust the phases of the two non-zero coefficients c_{kk} and c_{k1} so that both coefficients are just $1/\sqrt{2}$. From now on, the state-vectors $|Y_k'\rangle$ and $|\Psi_k'\rangle$ chosen in this way will be denoted $U|Y_k\rangle$ and $U|\Psi_k\rangle$. As we have seen,

$$U\frac{1}{\sqrt{2}}\big[|\Psi_k\rangle + |\Psi_1\rangle\big] = U|Y_k\rangle = \frac{1}{\sqrt{2}}\big[U|\Psi_k\rangle + U|\Psi_1\rangle\big]. \tag{2.A.5}$$

However, it still remains to define $U|\Psi\rangle$ for general state-vectors $|\Psi\rangle$.

Now consider an arbitrary state-vector $|\Psi\rangle$ belonging to an arbitrary ray $\mathcal{R}$, and expand it in the $|\Psi_k\rangle$:

$$|\Psi\rangle = \sum_k C_k|\Psi_k\rangle. \tag{2.A.6}$$

Any state $|\Psi'\rangle$ that belongs to the transformed ray $T\mathcal{R}$ may similarly be expanded in the complete orthonormal set $U|\Psi_k\rangle$:

$$|\Psi'\rangle = \sum_k C_k' U|\Psi_k\rangle. \tag{2.A.7}$$

The equality of $|\langle\Psi_k|\Psi\rangle|^2$ and $|\langle U\Psi_k|\Psi'\rangle|^2$ tells us that for all k (including $k=1$):

$$|C_k|^2 = |C_k'|^2, \tag{2.A.8}$$

while the equality of $|\langle Y_k|\Psi\rangle|^2$ and $|\langle UY_k|\Psi'\rangle|^2$ tells us that for all $k \neq 1$:

$$|C_k + C_1|^2 = |C_k' + C_1'|^2. \tag{2.A.9}$$

The ratio of Eqs. (2.A.9) and (2.A.8) yields the formula

$$\mathrm{Re}(C_k/C_1) = \mathrm{Re}(C_k'/C_1') \tag{2.A.10}$$

which with Eq. (2.A.8) also requires

$$\mathrm{Im}(C_k/C_1) = \pm\,\mathrm{Im}(C_k'/C_1') \tag{2.A.11}$$

and therefore either

$$C_k/C_1 = C_k'/C_1', \tag{2.A.12}$$

or else

$$C_k/C_1 = (C_k'/C_1')^*. \tag{2.A.13}$$

Furthermore, we can show that the same choice must be made for each k. (This step in the proof was omitted by Wigner.) To see this, suppose that for some k, we have

$C_k/C_1 = C'_k/C'_1$, while for some $l \neq k$, we have instead $C_l/C_1 = (C'_l/C'_1)^*$. Suppose also that both ratios are complex, so that these are really different cases. (This incidentally requires that $k \neq 1$ and $l \neq 1$, as well as $k \neq l$.) We will show that this is impossible.

Define a state-vector $|\Phi\rangle = \frac{1}{\sqrt{3}}[|\Psi_1\rangle + |\Psi_k\rangle + |\Psi_l\rangle]$. Since all the ratios of the coefficients in this state-vector are real, we must get the same ratios in any state-vector $|\Phi'\rangle$ belonging to the transformed ray:

$$|\Phi'\rangle = \frac{\alpha}{\sqrt{3}}\big[U|\Psi_1\rangle + U|\Psi_k\rangle + U|\Psi_l\rangle\big],$$

where α is a phase factor with $|\alpha| = 1$. But then the equality of the transition probabilities $|\langle\Phi|\Psi\rangle|$ and $|\langle\Phi'|\Psi'\rangle|$ requires that

$$\left|1 + \frac{C'_k}{C'_1} + \frac{C'_l}{C'_1}\right|^2 = \left|1 + \frac{C_k}{C_1} + \frac{C_l}{C_1}\right|^2$$

and hence

$$\left|1 + \frac{C_k}{C_1} + \frac{C^*_l}{C^*_1}\right|^2 = \left|1 + \frac{C_k}{C_1} + \frac{C_l}{C_1}\right|^2 .$$

This is only possible if

$$\mathrm{Re}\left(\frac{C_k C^*_l}{C_1 C^*_1}\right) = \mathrm{Re}\left(\frac{C_k C_l}{C_1 C_1}\right)$$

or, in other words, if

$$\mathrm{Im}\left(\frac{C_k}{C_1}\right)\mathrm{Im}\left(\frac{C_l}{C_1}\right) = 0.$$

Hence either C_k/C_1 or C_l/C_1 must be real for any pair k, l, in contradiction with our assumptions. We see then that for a given symmetry transformation T applied to a given state-vector $\sum_k C_k|\Psi_k\rangle$, we must have either Eq. (2.A.12) for all k, or else Eq. (2.A.13) for all k.

Wigner ruled out the second possibility, Eq. (2.A.13), because as he showed any symmetry transformation for which this possibility is realized would have to involve a reversal in the time coordinate, and in the proof he presented he was considering only symmetries like rotations that do not affect the direction of time. Here we are treating symmetries involving time-reversal on the same basis as all other symmetries, so we will have to consider that, for each symmetry T and state-vector $\sum_k C_k|\Psi_k\rangle$, either Eq. (2.A.12) or Eq. (2.A.13) may apply. Depending on which of these alternatives is realized, we will now define $U|\Psi\rangle$ to be the particular one of the state-vectors $|\Psi'\rangle$ belonging to the ray $T\mathcal{R}$ with phase chosen so that either $C_1 = C'_1$ or $C_1 = C'^*_1$, respectively. Then either

$$U\left(\sum_k C_k|\Psi_k\rangle\right) = \sum_k C_k U|\Psi_k\rangle \tag{2.A.14}$$

or else

$$U\left(\sum_k C_k|\Psi_k\rangle\right) = \sum_k C^*_k U|\Psi_k\rangle. \tag{2.A.15}$$

It remains to be proved that for a given symmetry transformation, we must make the same choice between Eqs. (2.A.14) and (2.A.15) for arbitrary values of the coefficients C_k. Suppose that Eq. (2.A.14) applies for a state-vector $\sum_k A_k|\Psi_k\rangle$ while Eq. (2.A.15) applies for a state-vector $\sum_k B_k|\Psi_k\rangle$. Then the invariance of transition probabilities requires that

$$\left|\sum_k B_k^* A_k\right|^2 = \left|\sum_k B_k A_k\right|^2$$

or equivalently

$$\sum_{kl} \mathrm{Im}(A_k^* A_l)\,\mathrm{Im}(B_k^* B_l) = 0. \tag{2.A.16}$$

We cannot rule out the possibility that Eq. (2.A.16) may be satisfied for a pair of state-vectors $\sum_k A_k|\Psi_k\rangle$ and $\sum_k B_k|\Psi_k\rangle$ belonging to different rays. However, for any pair of such state-vectors, with neither A_k nor B_k all of the same phase (so that Eqs. (2.A.14) and (2.A.15) are not the same), we can always find a third state-vector $\sum_k C_k|\Psi_k\rangle$ for which[10]

$$\sum_{kl} \mathrm{Im}(C_k^* C_l)\,\mathrm{Im}(A_k^* A_l) \neq 0 \tag{2.A.17}$$

and also

$$\sum_{kl} \mathrm{Im}(C_k^* C_l)\,\mathrm{Im}(B_k^* B_l) \neq 0. \tag{2.A.18}$$

As we have seen, it follows from Eq. (2.A.17) that the same choice between Eqs. (2.A.14) and (2.A.15) must be made for $\sum_k A_k|\Psi_k\rangle$ and $\sum_k C_k|\Psi_k\rangle$, and it follows from Eq. (2.A.18) that the same choice between Eqs. (2.A.14) and (2.A.15) must be made for $\sum_k B_k|\Psi_k\rangle$ and $\sum_k C_k|\Psi_k\rangle$, so the same choice between Eqs. (2.A.14) and (2.A.15) must also be made for the two state-vectors $\sum_k A_k|\Psi_k\rangle$ and $\sum_k B_k|\Psi_k\rangle$ with which we started. We have thus shown that for a given symmetry transformation T either all state-vectors satisfy Eq. (2.A.14) or else they all satisfy Eq. (2.A.15).

It is now easy to show that as we have defined it, the quantum mechanical operator U is either linear and unitary or antilinear and antiunitary. First, suppose that Eq. (2.A.14) is satisfied for all state-vectors $\sum_k C_k|\Psi_k\rangle$. Any two state-vectors $|\Psi\rangle$ and $|\Phi\rangle$ may be expanded as

$$|\Psi\rangle = \sum_k A_k|\Psi_k\rangle, \qquad |\Phi\rangle = \sum_k B_k|\Psi_k\rangle$$

[10]If for some pair k,l both $A_k^* A_l$ and $B_k^* B_l$ are complex, then choose all C's to vanish except for C_k and C_l, and choose these two coefficients to have different phases. If $A_k^* A_l$ is complex but $B_k^* B_l$ is real for some pair k,l, then there must be some other pair m,n (possibly with either m or n but not both equal to k or l) for which $B_m^* B_n$ is complex. If also $A_m^* A_n$ is complex, then choose all C's to vanish except for C_m and C_n, and choose these two coefficients to have different phase. If $A_m^* A_n$ is real, then choose all C's to vanish except for C_k, C_l, C_m, and C_n, and choose these four coefficients all to have different phases. The case where $B_k^* B_l$ is complex but $A_k^* A_l$ is real is handled in just the same way.

and so, using Eq. (2.A.14),

$$
\begin{aligned}
U\big(\alpha|\Psi\rangle + \beta|\Phi\rangle\big) &= U\sum_k(\alpha A_k + \beta B_k)|\Psi_k\rangle = \sum_k(\alpha A_k + \beta B_k)U|\Psi_k\rangle \\
&= \alpha\sum_k A_k U|\Psi_k\rangle + \beta\sum_k B_k U|\Psi_k\rangle.
\end{aligned}
$$

Using Eq. (2.A.14) again, this gives

$$
U\big(\alpha|\Psi\rangle + \beta|\Phi\rangle\big) = \alpha U|\Psi\rangle + \beta U|\Phi\rangle, \tag{2.A.19}
$$

so U is *linear.* Also, using Eqs. (2.A.2) and (2.A.3), the scalar product of the transformed states is

$$
\langle U\Psi|U\Phi\rangle = \sum_{kl} A_k^* B_l \langle U\Psi_k|U\Psi_l\rangle = \sum_k A_k^* B_k,
$$

and hence

$$
\langle U\Psi|U\Phi\rangle = \langle\Psi|\Phi\rangle, \tag{2.A.20}
$$

so U is *unitary.*

The case of a symmetry that satisfies Eq. (2.A.15) for all state-vectors may be dealt with in much the same way. The reader can probably supply the arguments without help, but since antilinear operators may be unfamiliar, we shall give the details here anyway. Suppose that Eq. (2.A.15) is satisfied for all state-vectors $\sum_k C_k|\Psi_k\rangle$. Any two state-vectors $|\Psi\rangle$ and $|\Phi\rangle$ may be expanded as before, and so:

$$
\begin{aligned}
U\big(\alpha|\Psi\rangle + \beta|\Phi\rangle\big) &= U\sum_k(\alpha A_k + \beta B_k)|\Psi_k\rangle \\
&= \sum_k(\alpha^* A_k^* + \beta^* B_k^*)U|\Psi_k\rangle \\
&= \alpha^*\sum_k A_k^* U|\Psi_k\rangle + \beta^*\sum_k B_k^* U|\Psi_k\rangle.
\end{aligned}
$$

Using Eq. (2.A.15) again, this gives

$$
U\big(\alpha|\Psi\rangle + \beta|\Phi\rangle\big) = \alpha^* U|\Psi\rangle + \beta^* U|\Phi\rangle, \tag{2.A.21}
$$

so U is *antilinear.* Also, using Eqs. (2.A.2) and (2.A.3), the scalar product of the transformed states is

$$
\langle U\Psi|U\Phi\rangle = \sum_{kl} A_k B_l^* \langle U\Psi_k|U\Psi_l\rangle = \sum_k A_k B_k^*,
$$

and hence

$$
\langle U\Psi|U\Phi\rangle = \langle\Psi|\Phi\rangle^*, \tag{2.A.22}
$$

so U is *antiunitary.*

Appendix B Group Operators and Homotopy Classes

In this appendix we shall prove the theorem stated in Section 2.7, that the phases of the operators $U(T)$ for finite symmetry transformations T may be chosen so that these operators form a representation of the symmetry group, rather than a projective representation, provided (a) the generators of the group can be defined so that there are no central charges in the Lie algebra, and (b) the group is simply connected. We shall also comment on the projective representations encountered for groups that are not simply connected, and their relation to the homotopy classes of the group.

To prove this theorem, let us recall the method by which we construct the operators corresponding to symmetry transformations. As described in Section 2.2, we introduce a set of real variables θ^a to parameterize these transformations, in such a way that the transformations satisfy the composition rule (2.2.15):

$$T(\bar{\theta})T(\theta) = T\big(f(\bar{\theta},\theta)\big).$$

We want to construct operators $U(T(\theta)) \equiv U[\theta]$ that satisfy the corresponding condition[11]

$$U[\bar{\theta}]U[\theta] = U\big[f(\bar{\theta},\theta)\big]. \tag{2.B.1}$$

To do this, we lay down arbitrary 'standard' paths $\Theta^a_\theta(s)$ in group parameter space, running from the origin to each point θ, with $\Theta^a_\theta(0) = 0$ and $\Theta^a_\theta(1) = \theta^a$, and define $U_\theta(s)$ along each such path by the differential equation

$$\frac{d}{ds}U_\theta(s) = it_a U_\theta(s) h^a{}_b\big(\Theta_\theta(s)\big)\frac{d\Theta^b_\theta(s)}{ds} \tag{2.B.2}$$

with the initial condition

$$U_\theta(0) = \mathbf{1}, \tag{2.B.3}$$

where

$$[h^{-1}]^a{}_b(\theta) \equiv \left[\frac{\partial f^a(\bar{\theta},\theta)}{\partial \bar{\theta}^b}\right]_{\bar{\theta}=0}. \tag{2.B.4}$$

We are eventually going to identify the operators $U[\theta]$ with $U_\theta(1)$, but first we must establish some of the properties of $U_\theta(s)$.

In order to check the composition rule, consider two points θ_1 and θ_2, and define a path γ that runs from 0 to θ_1 and thence to $f(\theta_2,\theta_1)$:

$$\Theta^a_\gamma(s) \equiv \begin{cases} \Theta^a_{\theta_1}(2s), & 0 \le s \le \frac{1}{2}, \\ f^a\big(\Theta_{\theta_2}(2s-1),\theta_1\big), & \frac{1}{2} \le s \le 1. \end{cases} \tag{2.B.5}$$

At the end of the first segment, we are at $U_\gamma(\frac{1}{2}) = U_{\theta_1}(1)$. To evaluate $U_\gamma(s)$ along the second segment, we need the derivative of $f^a(\Theta_{\theta_2}(2s-1),\theta_1)$. For this purpose, we use the fundamental associativity condition:

$$f^a\big(f(\theta_3,\theta_2),\theta_1\big) = f^a\big(\theta_3, f(\theta_2,\theta_1)\big). \tag{2.B.6}$$

[11] Square brackets are used here to distinguish U operators constructed as functions of the group parameters from those expressed as functions of the group transformations themselves.

Matching the coefficients of θ_3^c in the limit $\theta_3 \to 0$ yields the result:

$$\frac{\partial f^a(\theta_2,\theta_1)}{\partial\theta_2^b} h^b{}_c\big(f(\theta_2,\theta_1)\big) = h^a{}_c(\theta_2). \tag{2.B.7}$$

Along the second segment, the differential equation (2.B.2) for $U_\gamma(s)$ is thus the same as the differential equation for $U_{\theta_2}(2s-1)$. They satisfy different initial conditions, but $U_\gamma(s)U_{\theta_1}^{-1}(1)$ also satisfies the same differential equation as $U_{\theta_2}(2s-1)$, and in addition the same initial condition: at $s = \frac{1}{2}$, both are unity. We therefore conclude that for $\frac{1}{2} \le s \le 1$,

$$U_\gamma(s)U_{\theta_1}^{-1}(1) = U_{\theta_2}(2s-1)$$

and in particular

$$U_\gamma(1) = U_{\theta_2}(1)U_{\theta_1}(1). \tag{2.B.8}$$

However, this does *not* say that $U_\theta(1)$ satisfies the desired composition rule (2.B.1), because although the path $\Theta_\gamma(s)$ runs from $\theta^a = 0$ to $\theta^a = f^a(\theta_2,\theta_1)$, in general it will not be the same as whatever 'standard' path $\Theta_{f(\theta_2,\theta_1)}$ we have chosen to run directly from $\theta^a = 0$ to $\theta^a = f^a(\theta_2,\theta_1)$. We need to show that $U_\theta(1)$ is independent of the path from 0 to θ in order to be able to identify $U[\theta]$ as $U_\theta(1)$.

For this purpose, consider the variation δU of $U_\theta(s)$ produced by a variation $\delta\Theta(s)$ in the path from 0 to θ. Taking the variation of Eq. (2.B.2) gives the differential equation

$$\begin{aligned}\frac{d}{ds}\delta U = it_a\delta U\, h^a{}_b(\Theta)\frac{d\Theta^b}{ds} + it_aUh^a{}_{b,c}(\Theta)\delta\Theta^c\frac{d\Theta^b}{ds}\\ + it_aUh^a{}_b(\Theta)\frac{d\delta\Theta^b}{ds},\end{aligned}$$

where $h^a{}_{b,c} \equiv \partial h^a{}_b/\partial\theta^c$. Using the Lie commutation relations (2.2.22) (without central charges) and rearranging a bit, this gives

$$\begin{aligned}\frac{d}{ds}\big(U^{-1}\delta U\big) &= \frac{d}{ds}\big(iU^{-1}t_aUh^a{}_b\delta\Theta^b\big)\\ &+ iU^{-1}t_aU\delta\Theta^b\frac{d\Theta^c}{ds}\left(h^a{}_{c,b} - h^a{}_{b,c} + C^a{}_{ed}h^e{}_bh^d{}_c\right).\end{aligned} \tag{2.B.9}$$

However, by taking the limit $\theta_3, \theta_2 \to 0$ in the associativity condition (2.B.6), we find for all θ:

$$h(\theta)^a{}_{b,c} = -f^a{}_{de}h(\theta)^d{}_bh(\theta)^e{}_c, \tag{2.B.10}$$

where $f^a{}_{de}$ is the coefficient defined by (2.2.19). Antisymmetrizing in b and c shows that the last term in Eq. (2.B.9) vanishes

$$h^a{}_{c,b} - h^a{}_{b,c} + C^a{}_{ed}h^e{}_bh^d{}_c = 0. \tag{2.B.11}$$

Eq. (2.B.9) thus tells us that the quantity

$$U^{-1}\delta U - iU^{-1}t_aUh^a{}_b\delta\Theta^b$$

is constant along the path $\theta(s)$. It follows that $U_\theta(1)$ is stationary under any infinitesimal variation of the path that leaves the endpoints $\Theta(0) = 0$ and $\Theta(1) = \theta$ (and $U_\theta(0) = \mathbf{1}$) fixed. But assumption (b) tells us that any path from $\Theta(0) = 0$ to $\Theta(1) = \theta$ can be continuously deformed into any other, so we may now regard $U_\theta(1)$ as a path-independent function of θ alone:

$$U_\theta(1) \equiv U[\theta]. \tag{2.B.12}$$

In particular, since the path γ leads from 0 to $f(\theta_2, \theta_1)$, we have

$$U_\gamma(1) = U\big[f(\theta_2, \theta_1)\big] \tag{2.B.13}$$

so that Eq. (2.B.8) shows that $U[\theta]$ satisfies the group multiplication law (2.B.1), as was to be proved.

Now that we have constructed a non-projective representation $U[\theta]$, it remains to prove that any projective representation $\widetilde{U}[\theta]$ of the same group with the same representation generators t_a can only differ from $U[\theta]$ by a phase:

$$\widetilde{U}[\theta] = e^{i\alpha(\theta)} U[\theta]$$

so that the phase ϕ in the multiplication law for $\widetilde{U}[\theta]$:

$$\widetilde{U}[\theta']\widetilde{U}[\theta] = e^{i\phi(\theta',\theta)}\widetilde{U}\big[f(\theta', \theta)\big]$$

can be removed by a simple change of phase of $\widetilde{U}[\theta]$. To see this, consider the operator

$$U[\theta]^{-1}U[\theta']^{-1}\widetilde{U}[\theta']\widetilde{U}[\theta] = U\big[f(\theta', \theta)\big]^{-1}\widetilde{U}\big[f(\theta', \theta)\big]e^{i\phi(\theta',\theta)}.$$

Because $U[\theta]$ and $\widetilde{U}[\theta]$ have the same generators, the derivative of the left-hand side with respect to θ'^a vanishes at $\theta' = 0$, and so

$$0 = \frac{\partial}{\partial\theta^b}\Big\{U[\theta]^{-1}\widetilde{U}[\theta]\Big\} + i\phi_b(\theta)U[\theta]^{-1}\widetilde{U}[\theta],$$

where

$$\phi_b(\theta) \equiv h^a{}_b(\theta)\left[\frac{\partial}{\partial\theta'^a}\phi(\theta', \theta)\right]_{\theta'=0}.$$

Differentiating this result with respect to θ^c and antisymmetrizing in b and c gives immediately

$$0 = \frac{\partial\phi_b(\theta)}{\partial\theta^c} - \frac{\partial\phi_c(\theta)}{\partial\theta^b}.$$

A familiar theorem [13] tells us that in a simply connected space, this requires that ϕ_b is just a gradient of some function β:

$$\phi_b(\theta) = \frac{\partial\beta(\theta)}{\partial\theta^b}.$$

Thus the quantity $U[\theta]^{-1}\widetilde{U}[\theta]e^{i\beta(\theta)}$ is actually constant in θ. Setting it equal to its value at $\theta = 0$, we see that $\widetilde{U}$ is just proportional to U:

$$\widetilde{U}[\theta] = U[\theta]\exp\big(-i\beta(\theta) + i\beta(0)\big)$$

as claimed above.

* * *

The above analysis provides some information about the nature of the phase factors that can appear in the group multiplication law when the Lie algebra is free of central charges but the group is not simply connected. Suppose that the path γ from zero to θ to $f(\bar{\theta}, \theta)$ cannot be deformed into the standard path we have chosen to go from zero to $f(\bar{\theta}, \theta)$, or in other words, that the loop from zero to θ to $f(\bar{\theta}, \theta)$ and then back to zero is not continuously deformable to a point. Then $U[f(\theta_2, \theta_1)]^{-1}U[\theta_2]U[\theta_1]$ can be a phase factor $\exp(i\phi(\theta_2, \theta_1)) \neq \mathbf{1}$, but ϕ will be the same for all other loops into which this can be continuously deformed. The set consisting of all loops that start and end at the origin and that can be continuously deformed into a given loop is known as the *homotopy class* [14] of that loop; we have thus seen that $\phi(\theta_2, \theta_1)$ depends only on the homotopy class of the loop from zero to θ to $f(\bar{\theta}, \theta)$ and then back to zero. The set of homotopy classes forms a group; the 'product' of the homotopy class for loops $\mathcal{L}_1$ and $\mathcal{L}_2$ is the homotopy class of the loop formed by going around $\mathcal{L}_1$ and then $\mathcal{L}_2$; the 'inverse' of the homotopy class of the loop $\mathcal{L}$ is the homotopy class of the loop obtained by going around $\mathcal{L}$ in the opposite direction; and the 'identity' is the homotopy class of loops that can be deformed into a point at the origin. This group is known as the *first homotopy group* or *fundamental group* of the space in question. It is easy to see that the phase factors form a representation of this group: if going around loop $\mathcal{L}$ gives a phase factor $e^{i\phi}$, and going around loop $\bar{\mathcal{L}}$ gives a phase factor $e^{i\bar{\phi}}$, then going around both loops gives a phase factor $e^{i\phi}e^{i\bar{\phi}}$. Hence we can catalog all the possible types of projective representations of a given group $\mathcal{G}$ (with no central charges) if we know the one-dimensional representations of the first homotopy group of the parameter space of $\mathcal{G}$. Homotopy groups will be discussed in greater detail in Volume II.

Appendix C Inversions and Degenerate Multiplets

It is usually assumed that the inversions T and P take one-particle states into other one-particle states of the same species, perhaps with phase factors that depend on the particle species. In Section 2.6 we noted in passing that inversions might act in a more complicated way than this on degenerate multiplets of one-particle states, a possibility that seems to have been first suggested by Wigner [15] in 1964. This appendix will explore generalized versions of the inversion operators, in which finite matrices appear in place of the inversion phases, but without making some of Wigner's limiting assumptions.

Let us start with time-reversal. Wigner limited the possible action of the inversion operators by assuming that their squares are proportional to the unit operator. Because T is antiunitary, it is easy to see that the corresponding proportionality factor for T^2 can only be ± 1, perhaps with different signs for subspaces separated by superselection rules. When the sign for T^2 on the space of states with even or odd values of $2j$ is opposite to the sign $(-1)^{2j}$ found in Section 2.6, the physical states involved must furnish representations of the operator T that are more complicated than that assumed so far. But if we are willing to admit this possibility, there does not seem to be any good reason to impose Wigner's condition that T^2 is proportional to unity. It is not convincing to appeal to the structure of the extended Poincaré group; the only useful definition of any of the inversion operators is one that makes the operator exactly or approximately conserved, and this may not be the definition that makes T^2 proportional to the unit operator.

To explore more general possibilities for time-reversal, let us assume that on a massive one-particle state it has the action

$$T|\mathbf{p},\sigma,n\rangle = (-1)^{j-\sigma}\sum_m \Theta_{mn}|-\mathbf{p},-\sigma,m\rangle, \tag{2.C.1}$$

where $\mathbf{p}$, j, and σ are the particle's momentum, spin, and spin z-component, and n, m are indices labelling members of a degenerate multiplet of particle species. (The appearance of the factor $(-1)^{j-\sigma}$ and the reversal of $\mathbf{p}$ and σ are deduced in the same way as in Section 2.6.) The matrix Θ_{mn} is unknown, except that because T is antiunitary, Θ must be unitary.

Now let us see how we can simplify this transformation by an appropriate choice of basis for the one-particle states. Defining new states by the unitary transformation $|\mathbf{p},\sigma,n\rangle' = \sum_m \mathcal{U}_{mn}|\mathbf{p},\sigma,m\rangle$, we find the same transformation (2.C.1), with the matrix Θ_{mn} changed to

$$\Theta' = \mathcal{U}^{-1}\Theta\mathcal{U}^*. \tag{2.C.2}$$

We cannot in general make Θ' diagonal by such a choice of basis of the one-particle states, as we could if T were unitary. But we can instead make it block-diagonal, with the blocks either 1×1 phases, or 2×2 matrices of the form

$$\begin{pmatrix} 0 & e^{i\phi/2} \\ e^{-i\phi/2} & 0 \end{pmatrix}, \tag{2.C.3}$$

where the ϕ are various real phases.

(Here is the proof. First, note that Eq. (2.C.2) gives

$$\Theta'\Theta'^* = \mathcal{U}^{-1}\Theta\Theta^*\mathcal{U}.$$

This is a unitary transformation, so it can be chosen to diagonalize the unitary matrix $\Theta\Theta^*$. Assuming this to have been done, and dropping primes, we have

$$\Theta = D\Theta^T, \tag{2.C.4}$$

where D is a unitary diagonal matrix, say with phases $e^{i\phi_n}$ along the main diagonal. One immediate consequence is that the diagonal component Θ_{nn} vanishes unless $e^{i\phi_n} = 1$. Furthermore, if $e^{i\phi_n} = 1$ but $e^{i\phi_m} \neq 1$, then Eq. (2.C.4) tells us that $\Theta_{nm} = \Theta_{mn} = 0$. By listing first all rows and columns for which $e^{i\phi_n} = 1$, the matrix Θ is put in the form

$$\Theta = \begin{pmatrix} \mathcal{A} & 0 \\ 0 & \mathcal{B} \end{pmatrix}, \tag{2.C.5}$$

where $\mathcal{A}$ is symmetric as well as unitary, and the diagonal elements of $\mathcal{B}$ all vanish. Because $\mathcal{A}$ is symmetric, it can be expressed as the exponential of a symmetric anti-Hermitian matrix, so it can be diagonalized by a transformation (2.C.2) acting only on $\mathcal{A}$, with the corresponding submatrix of $\mathcal{U}$ real and hence orthogonal. It is therefore only necessary to consider the submatrix $\mathcal{B}$ that connects the rows and columns for which $e^{i\phi_n} \neq 0$. For $n \neq m$, Eq. (2.C.4) gives $\Theta_{nm} = e^{i\phi_n}\Theta_{mn}$ and $\Theta_{mn} = e^{i\phi_m}\Theta_{nm}$, so $\Theta_{nm} = e^{i\phi_n}e^{i\phi_m}\Theta_{nm}$ and also $\Theta_{mn} = e^{i\phi_n}e^{i\phi_m}\Theta_{mn}$. Hence $\Theta_{nm} = \Theta_{mn} = 0$ unless $e^{i\phi_n}e^{i\phi_m} = 1$. If we list first all rows and columns of $\mathcal{B}$ with a given phase $e^{i\phi_1} \neq 1$, and then all rows and columns with the opposite phase, and then all rows and columns with some other phase $e^{i\phi_2} \neq 1$ not equal to $e^{\pm i\phi_1}$, and then all rows and columns with opposite phase, and so on, the matrix $\mathcal{B}$ becomes of block diagonal form

$$\mathcal{B} = \begin{pmatrix} \mathcal{B}_1 & 0 & \cdots \\ 0 & \mathcal{B}_2 & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}, \tag{2.C.6}$$

where

$$\mathcal{B}_i = \begin{pmatrix} 0 & e^{i\phi_i/2}\mathcal{C}_i \\ e^{-i\phi_i/2}\mathcal{C}_i^T & 0 \end{pmatrix}. \tag{2.C.7}$$

Furthermore, the unitarity of Θ and hence of $\mathcal{B}$ requires that $\mathcal{C}_i\mathcal{C}_i^\dagger = \mathcal{C}_i^\dagger\mathcal{C}_i = \mathbf{1}$, and hence $\mathcal{C}_i$ is square and unitary. By applying a transformation (2.C.2) with $\mathcal{U}$ block-diagonal in the same sense as Θ, and with the matrix in the ith block of form

$$\begin{pmatrix} V_i & 0 \\ 0 & W_i \end{pmatrix}$$

with V_i and W_i unitary, the submatrices $\mathcal{C}_i$ are subjected to the transformations $\mathcal{C}_i \longrightarrow V_i^{-1}\mathcal{C}_iW_i^*$, so we can clearly choose this transformation to make $\mathcal{C}_i = \mathbf{1}$. This establishes a correspondence between pairs of individual rows and columns within each block with phases $e^{i\phi_i}$ and $e^{-i\phi_i}$. To put the matrix $\mathcal{B}$ into block-diagonal form with 2×2 blocks of form (2.C.3), it is now only necessary to rearrange the rows and columns so that within the ith block we list rows and columns with phase $e^{i\phi_i}$ alternating with the corresponding rows and columns with phase $e^{-i\phi_i}$.)

It is important to note that where $e^{i\phi} \neq 1$, it is not possible to choose states to diagonalize the time-reversal transformation. If we have a pair of states $|\mathbf{p}, \sigma, \pm\rangle$ on which T acts with a matrix (2.C.3), then

$$T|\mathbf{p}, \sigma, \pm\rangle = e^{\pm i\phi/2}(-1)^{j-\sigma}|-\mathbf{p}, -\sigma, \mp\rangle. \tag{2.C.8}$$

Then on an arbitrary linear combination of these states, time-reversal gives

$$T\big(c_+|\mathbf{p}, \sigma, +\rangle + c_-|\mathbf{p}, \sigma, -\rangle\big) = (-1)^{j-\sigma}\big(e^{i\phi/2}c_+^*|-\mathbf{p}, -\sigma, -\rangle + e^{-i\phi/2}c_-^*|-\mathbf{p}, -\sigma, +\rangle\big).$$

For $c_+|\mathbf{p}, \sigma, +\rangle + c_-|\mathbf{p}, \sigma, -\rangle$ to be transformed under T by a phase λ, it is necessary that

$$e^{i\phi/2}c_+^* = \lambda c_-, \qquad e^{-i\phi/2}c_-^* = \lambda c_+.$$

But combining these equations gives $e^{\pm i\phi/2}c_\pm^* = |\lambda|^2 c_\pm^* e^{\mp i\phi/2}$, which is impossible unless either $c_+ = c_- = 0$ or $e^{i\phi}$ is unity. Thus for $e^{i\phi} \neq 1$, time-reversal invariance imposes a two-fold degeneracy on these states, beyond that associated with their spin.

Of course, if there is an additional 'internal' symmetry operator S which subjects these states to the transformation

$$S|\mathbf{p},\sigma,\pm\rangle = e^{\pm i\phi/2}|-\mathbf{p},\sigma,\mp\rangle,$$

then we can redefine the time-reversal operator as $T' \equiv S^{-1}T$, and this operator would not mix the states $|\mathbf{p},\sigma,\pm\rangle$ with one another. It is only in the case where no such internal symmetry exists that we can attribute the doubling of particle states to time-reversal itself.

Let's come back now to the question of the square of T. Repeating the transformation (2.C.8) gives

$$T^2|\mathbf{p},\sigma,\pm\rangle = (-1)^{2j}e^{\mp i\phi}|\mathbf{p},\sigma,\pm\rangle. \tag{2.C.9}$$

If we were to assume with Wigner that T^2 is proportional to the unit operator, then we would have to have $e^{i\phi} = e^{-i\phi}$, and since the phase is then real it would have to be $+1$ or -1. The choice $e^{i\phi} = -1$ would still require a two-fold degeneracy of one-particle states beyond that associated with their spin, and under Wigner's assumptions *all* particles would show this doubling. But there is no reason not to take a general phase ϕ in Eq. (2.C.8), one that may vanish for some particles and not for others. Thus the fact that observed particles do not show the extra two-fold degeneracy does not rule out the possibility that others might.

We may also consider the possibility of more complicated representations of the parity operator P, with

$$P|\mathbf{p},\sigma,n\rangle = \sum_m \Pi_{mn}|-\mathbf{p},\sigma,m\rangle \tag{2.C.10}$$

with a unitary but otherwise unconstrained matrix Π. Unlike the case of time-reversal, here we may always diagonalize this matrix by a choice of basis for the states. But this choice of basis may not be the one in which time-reversal acts simply, so, in principle, P and T together can impose additional degeneracies that would not be required by P or T alone.

As discussed in Chapter 5, any quantum field theory is expected to respect a symmetry known as CPT, which acts on one-particle states as

$$CPT|\mathbf{p},\sigma,n\rangle = (-1)^{j-\sigma}|\mathbf{p},-\sigma,n^c\rangle, \tag{2.C.11}$$

where n^c denotes the antiparticle (or 'charge-conjugate') of particle n. No phases or matrices are allowed in this transformation (though of course we could always introduce such phases or matrices by combining CPT with good internal symmetries.) It follows that

$$(CPT)^2|\mathbf{p},\sigma,n\rangle = (-1)^{2j}|\mathbf{p},-\sigma,n\rangle, \tag{2.C.12}$$

so the possibility suggested by Wigner of a sign $-(-1)^{2j}$ in the action of $(CPT)^2$ does not arise in quantum field theory.

To the extent that T is a good symmetry of some class of phenomena, so is the inversion $CP \equiv (CPT)T^{-1}$. For the states that transform under T in the conventional way

$$T|\mathbf{p},\sigma,n\rangle \propto |-\mathbf{p},-\sigma,n\rangle, \tag{2.C.13}$$

the CP operator also acts conventionally

$$CP|\mathbf{p},\sigma,n\rangle \propto |-\mathbf{p},\sigma,n^c\rangle. \tag{2.C.14}$$

The operator $C \equiv CPP^{-1}$ then just interchanges particles and antiparticles

$$C|\mathbf{p},\sigma,n\rangle \propto |\mathbf{p},\sigma,n^c\rangle. \tag{2.C.15}$$

On the other hand, where T has the unconventional representation (2.C.8), Eq. (2.C.11) gives

$$CP|\mathbf{p},\sigma,\pm\rangle = e^{\mp i\phi/2}|-\mathbf{p},\sigma,\mp^c\rangle \tag{2.C.16}$$

In particular, it is possible that the degeneracy indicated by the label $\pm$ may be the same as the particle–antiparticle degeneracy, so that the antiparticle (as defined by CPT) of the state $|\Psi_\pm\rangle$ is $|\Psi_\mp\rangle$. In this case, CP would have the unconventional property of *not* interchanging particles and antiparticles. As far as these particles are concerned, CP and T would be what are usually called P and CT. But this is not merely a matter of definition; on other particles CP and T would still have their usual effect.

No examples are known of particles that furnish unconventional representations of inversions, so these possibilities will not be pursued further here. From now on, the inversions will be assumed to have the conventional action assumed in Section 2.6.

Problems

1. Suppose that observer $\mathcal{O}$ sees a W-boson (spin one and mass $m \neq 0$) with momentum $\mathbf{p}$ in the y-direction and spin z-component σ. A second observer $\mathcal{O}'$ moves relative to the first with velocity $\mathbf{v}$ in the z-direction. How does $\mathcal{O}'$ describe the W state?

2. Suppose that observer $\mathcal{O}$ sees a photon with momentum $\mathbf{p}$ in the y-direction and polarization vector in the z-direction. A second observer $\mathcal{O}'$ moves relative to the first with velocity $\mathbf{v}$ in the z-direction. How does $\mathcal{O}'$ describe the same photon?

3. Derive the commutation relations for the generators of the Galilean group directly from the group multiplication law (without using our results for the Lorentz group). Include the most general set of central charges that cannot be eliminated by redefinition of the group generators.

4. Show that the operators $P_\mu P^\mu$ and $W_\mu W^\mu$ commute with all Lorentz transformation operators $U(\Lambda, a)$, where $W_\mu \equiv \epsilon_{\mu\nu\rho\lambda}J^{\nu\rho}P^\lambda$.

5. Consider physics in two space and one time dimensions, assuming invariance under a 'Lorentz' group $SO(2,1)$. How would you describe the spin states of a single massive particle? How do they behave under Lorentz transformations? What about the inversions P and T?

6. As in Problem 5, consider physics in two space and one time dimensions, assuming invariance under a 'Lorentz' group $O(2,1)$. How would you describe the spin states of a single massless particle? How do they behave under Lorentz transformations? What about the inversions P and T?

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3 Scattering Theory

3.1 'In' and 'Out' States

3.2 The S-Matrix

3.3 Symmetries of the S-Matrix

Parity

To the extent that the symmetry under the transformation $\mathbf{x} \to -\mathbf{x}$ is really valid, there must exist a unitary operator P under which both 'in' and 'out' states transform as a direct product of single-particle states:

$$\mathsf{P}\,|p_1\sigma_1 n_1;\, p_2\sigma_2 n_2;\, \cdots\rangle^{\pm} = \eta_{n_1}\eta_{n_2}\cdots\,|\mathcal{P}p_1\sigma_1 n_1;\, \mathcal{P}p_2\sigma_2 n_2;\, \cdots\rangle^{\pm} \tag{3.3.41}$$

where η_n is the intrinsic parity of particles of species n, and $\mathcal{P}$ reverses the space components of p^μ. (This is for massive particles; the modification for massless particles is obvious.) The parity conservation condition for the S-matrix is then:

$$\begin{aligned} S_{p_1'\sigma_1'n_1';\,p_2'\sigma_2'n_2';\,\cdots\,,\,p_1\sigma_1n_1;\,p_2\sigma_2n_2;\,\cdots} &= \eta^*_{n_1'}\eta^*_{n_2'}\cdots\,\eta_{n_1}\eta_{n_2}\cdots \\ &\quad\times S_{\mathcal{P}p_1'\sigma_1'n_1';\,\mathcal{P}p_2'\sigma_2'n_2';\,\cdots\,,\,\mathcal{P}p_1\sigma_1n_1;\,\mathcal{P}p_2\sigma_2n_2;\,\cdots}\,. \end{aligned} \tag{3.3.42}$$

Just as for internal symmetries, an operator P satisfying Eq. (3.3.41) will actually exist if the operator P_0 which is defined to act this way on free-particle states commutes with H_{int} as well as H_0.

The phases η_n may be inferred either from dynamical models or from experiment, but neither can provide a unique determination of the ηs. This is because we are always free to redefine P by combining it with any conserved internal symmetry operator. For instance, if P is conserved, then so is

$$\mathsf{P}' \equiv \mathsf{P}\exp(i\alpha B + i\beta L + i\gamma Q),$$

where B, L, and Q are respectively baryon number, lepton number, and electric charge, and α, β, and γ are arbitrary real phases; hence either P or P' could be called the parity operator. The neutron, proton, and electron have different combinations of values for B, L, and Q, so by judicious choice of the phases α, β, and γ we can define the intrinsic parities of all three particles to be $+1$. However, once we have done this the intrinsic parities of other particles like the charged pion (which can be emitted in a transition $n \to p + \pi^-$) are no longer arbitrary. Also, the intrinsic parity of any particle like the neutral pion π^0 which carries no conserved quantum numbers is always meaningful.

The foregoing remarks help to clarify the question of whether intrinsic parities must always have the values ± 1. It is easy to say that space inversion P has the group multiplication law $\mathsf{P}^2 = 1$; however, the parity operator that is conserved may not be this one, but rather may differ from it by a phase transformation of some sort. In any case, whether or not $\mathsf{P}^2 = 1$, the operator P^2 behaves just like an internal symmetry transformation:

$$\mathsf{P}^2|p_1\sigma_1 n_1;\, p_2\sigma_2 n_2;\, \cdots\rangle^{\pm} = \eta^2_{n_1}\eta^2_{n_2}\cdots\,|p_1\sigma_1 n_1;\, p_2\sigma_2 n_2;\, \cdots\rangle^{\pm}\,.$$

If this internal symmetry is part of a continuous symmetry group of phase transformations, such as the group of multiplication by the phases $\exp(i\alpha B + i\beta L + i\gamma Q)$ with arbitrary values of α, β, and γ, then its inverse square root must also be a member of this group, say I_P, with $I_P^2\,\mathsf{P}^2 = 1$ and $[I_P, \mathsf{P}] = 0$. (For instance, if $\mathsf{P}^2 = \exp(i\alpha B + \cdots)$, then take $I_P = \exp(-\frac{1}{2}i\alpha B + \cdots)$.) We can then define a new parity operator $\mathsf{P}' \equiv \mathsf{P} I_P$ with $\mathsf{P}'^2 = 1$. This is conserved to the same extent as P, so there is no reason why we should not call this *the* parity operator, in which case the intrinsic parities can only take the values ± 1.

The only sort of theory in which it is not necessarily possible to define parity so that all intrinsic parities have the values ± 1 is one in which there is some discrete internal symmetry which is not a member of any continuous symmetry group of phase transformations.[10] For instance, it is a consequence of angular-momentum conservation that the total number F of all particles of half-integer spin can only change by even numbers, so the internal symmetry operator $(-1)^F$ is conserved. All known particles of half-integer spin have odd values of the sum $B+L$ of baryon number and lepton number, so as far as we know, $(-1)^F = (-1)^{B+L}$. If this is true, then $(-1)^F$ is part of a continuous symmetry group, consisting of the operators $\exp(i\alpha(B+L))$ with arbitrary real α, and has an inverse square root $\exp(-i\pi(B+L)/2)$. In this case, if $\mathsf{P}^2 = (-1)^F$ then P can be redefined so that all intrinsic parities are ± 1. However, if there were to be discovered a particle of half-integer spin and an even value of $B+L$ (such as a so-called Majorana neutrino, with $j = \frac{1}{2}$ and $B+L = 0$), then it would be possible to have $\mathsf{P}^2 = (-1)^F$ without our being able to redefine the parity operator itself to have eigenvalues ± 1. In this case, of course, we would have $\mathsf{P}^4 = 1$, so all particles would have intrinsic parities either ± 1 or (like the Majorana neutrino) $\pm i$.

It follows from Eq. (3.3.42) that if the product of intrinsic parities in the final state is equal to the product of intrinsic parities in the initial state, or equal to minus this product, then the S-matrix must be respectively even or odd overall in the three-momenta. For instance, it was observed[11] in 1951 that a pion can be absorbed by a deuteron from the $\ell = 0$ ground state of the $\pi^- d$ atom, in the reaction $\pi^- + d \to n + n$. (As discussed in Section 3.7, the orbital angular-momentum quantum number ℓ can be used in relativistic physics in the same way as in non-relativistic wave mechanics.) The initial state has total angular-momentum $j = 1$ (the pion and deuteron having spins zero and one, respectively), so the final state must have orbital angular-momentum $\ell = 1$ and total neutron spin $s = 1$. (The other possibilities $\ell = 1$, $s = 0$; $\ell = 0$, $s = 1$; and $\ell = 2$, $s = 1$, which are allowed by angular-momentum conservation, are forbidden by the requirement that the final state be antisymmetric in the two neutrons.) Because the final state has $\ell = 1$, the matrix element is odd under reversal of the direction of all three-momenta, so we can conclude that the intrinsic parities of the particles in this reaction must be related by:

$$\eta_d \eta_{\pi^-} = -\eta_n^2\,.$$

The deuteron is known to be a bound state of a proton and neutron with even orbital angular-momentum (chiefly $\ell = 0$), and as we have seen we can take the neutron and proton to have the same intrinsic parity, so $\eta_d = \eta_n^2$, and we can conclude that $\eta_{\pi^-} = -1$; that is, the negative pion is a *pseudoscalar* particle. The π^+ and π^0 have also been found to have negative parity, as would be expected from the symmetry (isospin invariance) among these three particles.

The negative parity of the pion has some striking consequences. A spin zero particle that decays into three pions must have intrinsic parity $\eta_\pi^3 = -1$, because in the Lorentz frame in which the decaying particle is at rest, rotational invariance only allows the matrix element to depend on scalar products of the pion momenta with each other, all of which are even under reversal of all momenta. (The triple scalar product $\mathbf{p}_1 \cdot (\mathbf{p}_2 \times \mathbf{p}_3)$ formed from the three pion momenta vanishes because $\mathbf{p}_1 + \mathbf{p}_2 + \mathbf{p}_3 = 0$.) For the same reason, a spin zero particle that decays into two pions must have intrinsic parity $\eta_\pi^2 = +1$. In particular, among the strange particles discovered in the late 1940s there seemed to be two different particles of zero spin (inferred from the angular distribution of their decay products): one, the τ, was identified by its decay into three pions, and hence was assigned a parity -1, while the other, the θ, was identified by its decay into two pions and was assigned a parity $+1$. The trouble with all this was that as the τ and θ were studied in greater detail, they seemed increasingly to have identical masses and lifetimes. After many suggested solutions of this puzzle, Lee and Yang in 1956 finally cut the Gordian knot, and proposed that the τ and θ are the same particle, (now known as the $K^\pm$) and that parity is simply not conserved in the weak interactions that lead to its decay.[12]

As we shall see in detail in the next section of this chapter, the rate for a physical process $\alpha \to \beta$ (with $\alpha \neq \beta$) is proportional to $|S_{\beta\alpha}|^2$, with proportionality factors that are invariant under reversal of all three-momenta. As long as the states α and β contain definite numbers of particles of each type, the phase factors in Eq. (3.3.42) have no effect on $|S_{\beta\alpha}|^2$, so Eq. (3.3.42) would imply that the rate for $\alpha \to \beta$ is invariant under the reversal of direction of all three-momenta. As we have seen, this is a trivial consequence of rotational invariance for the decays of a K meson into two or three pions, but it is a non-trivial restriction on rates in more complicated processes. For example, following theoretical suggestions by Lee and Yang, Wu together with a group at the National Bureau of Standards measured the angular distribution of the electron in the final state of the beta decay $\mathrm{Co}^{60} \to \mathrm{Ni}^{60} + e^- + \bar{\nu}$ with a polarized cobalt source.[13] (No attempt was made in this experiment to measure the momentum of the antineutrino or nickel nucleus.) The electrons were found to be preferentially emitted in a direction opposite to that of the spin of the decaying nucleus, which would, of course, be impossible if the decay rate were invariant under a reversal of all three-momenta. A similar result was found in the decay of a positive muon (polarized in its production in the process $\pi^+ \to \mu^+ + \nu$) into a positron, neutrino, and antineutrino.[14] In this way, it became clear that parity is indeed not conserved in the weak interactions responsible for these decays. Nevertheless, for reasons discussed in Section 12.5, parity *is* conserved in the strong and electromagnetic interactions, and therefore continues to play an important part in theoretical physics.

3.4 Rates and Cross-Sections

3.5 Perturbation Theory

3.6 Implications of Unitarity

3.7 Partial-Wave Expansions*

3.8 Resonances*

4 The Cluster Decomposition Principle

4.1 Bosons and Fermions

4.2 Creation and Annihilation Operators

4.3 Cluster Decomposition and Connected Amplitudes

4.4 Structure of the Interaction

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5 Quantum Fields and Antiparticles

5.1 Free Fields

5.2 Causal Scalar Fields

5.3 Causal Vector Fields

5.4 The Dirac Formalism

5.5 Causal Dirac Fields

5.6 General Irreducible Representations of the Homogeneous Lorentz Group*

5.7 General Causal Fields*

5.8 The CPT Theorem

5.9 Massless Particle Fields

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6 The Feynman Rules

6.1 Derivation of the Rules

6.2 Calculation of the Propagator

6.3 Momentum Space Rules

6.4 Off the Mass Shell

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7 The Canonical Formalism

7.1 Canonical Variables

7.2 The Lagrangian Formalism

7.3 Global Symmetries

7.4 Lorentz Invariance

7.5 Transition to Interaction Picture: Examples

7.6 Constraints and Dirac Brackets

7.7 Field Redefinitions and Redundant Couplings*

Appendix Dirac Brackets from Canonical Commutators

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8 Electrodynamics

9 Path-Integral Methods

10 Non-Perturbative Methods

10.1 Symmetries

10.2 Polology

10.3 Field and Mass Renormalization

10.4 Renormalized Charge and Ward Identities

10.5 Gauge Invariance

10.6 Electromagnetic Form Factors and Magnetic Moment

10.7 The Källén–Lehmann Representation*

10.8 Dispersion Relations*

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11 One-Loop Radiative Corrections in Quantum Electrodynamics

11.1 Counterterms

11.2 Vacuum Polarization

11.3 Anomalous Magnetic Moments and Charge Radii

11.4 Electron Self-Energy

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12 General Renormalization Theory

12.1 Degrees of Divergence

12.2 Cancellation of Divergences

12.3 Is Renormalizability Necessary?

12.4 The Floating Cutoff*

12.5 Accidental Symmetries*

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13 Infrared Effects

14 Bound States in External Fields

14.1 The Dirac Equation

14.2 Radiative Corrections in External Fields

14.3 The Lamb Shift in Light Atoms

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References